

Telecom Customer Churn

Zara Asim 29262, Maryam Sultan 29186, Hadiya Muneeb 28424

1. Interviewee Details:

Interviewee 1:

Mr. Aman - Assistant Manager, Wireless - PTCL

Interviewee 2:

Mr. Husnain - Retired Engineering Supervisor – PTCL

They did not have LinkedIn Profiles.

2. Dataset Overview

2.1 Dataset Description

The dataset used in this analysis is the Telecom Customer Churn dataset published by Maven Analytics, available on Kaggle. It contains 7,043 customer records across 38 features. The overall churn rate in this dataset sits at approximately 26.5%.

2.2 Columns

- Identifier: CustomerID
- Demographics: Gender, Age, Married, Number of Dependents.
- Location: City, Zip Code, Latitude, Longitude.
- Account Info: Tenure in Months, Number of Referrals, Offer, Contract, Paperless Billing, Payment Method.
- Phone Services: Phone Service, Avg Monthly Long Distance Charges, Multiple Lines.
- Internet Services: Internet Service, Internet Type, Avg Monthly GB Download, Online Security, Online Backup, Device Protection Plan, Premium Tech Support, Streaming TV, Streaming Movies, Streaming Music, Unlimited Data.
- Financials: Monthly Charge, Total Charges, Total Refunds, Total Extra Data Charges, Total Long Distance Charges, Total Revenue.
- Churn Labels: Customer Status, Churn Category, Churn Reason.

2.3 Industry Background

The telecommunications industry is one of the most competitive markets in the world. With most consumers already owning mobile and internet subscriptions, operators can no longer grow by simply attracting new customers as they have to retain existing ones. Losing a customer is costly, as acquiring a new one can cost five to ten times more than keeping an existing one. Industry churn rates typically fall between 15% and 25% annually, and at 26.5%, our dataset exceeds that benchmark, making churn a pressing business problem.

3. Data Wrangling and EDA

3.1 Data Wrangling

Initial inspection revealed several missing values concentrated in service-related fields and churn-related fields. Blank strings and whitespace entries were first standardized into NaN values to ensure consistent handling of missing data.

A Python notebook using Pandas was used to:

- detect null values,
- identify blank or whitespace entries,
- verify whether missing values were logically valid,
- and ensure that service-related missing values matched the conditions described in the dataset documentation.

The dataset dictionary explicitly stated that certain fields are empty when customers do not subscribe to specific services. Therefore, missing values were not treated as random errors immediately. Instead, validation checks were performed, confirming that the missing values followed the expected business logic.

For customers where Customer Status was not Churned, the following fields were expected to be empty:

- Churn Category
- Churn Reason

For customers where Phone Service = No, the following fields were expected to be empty:

- Multiple Lines
- Avg Monthly Long Distance Charges

For customers where Internet Service = No, the following fields were expected to be empty:

- Internet Type
- Avg Monthly GB Download
- Online Security
- Online Backup
- Device Protection Plan
- Premium Tech Support
- Streaming TV
- Streaming Movies
- Streaming Music
- Unlimited Data

Since the missing values were logically caused by unavailable services rather than data corruption, they were safely replaced using Power BI Power Query transformations. The columns Churn Category and Churn Reason were left unchanged because they only apply to customers who churned.

The following table summarizes the transformations:

Column	Transformation Applied
Multiple Lines	Replaced null with “No”
Avg Monthly Long Distance Charges	Replaced null with 0
Internet Type	Replaced null with “None”
Avg Monthly GB Download	Replaced null with 0
Online Security	Replaced null with “No”
Online Backup	Replaced null with “No”
Device Protection Plan	Replaced null with “No”
Premium Tech Support	Replaced null with “No”
Streaming TV	Replaced null with “No”
Streaming Movies	Replaced null with “No”
Streaming Music	Replaced null with “No”
Unlimited Data	Replaced null with “No”

3.2 EDA

After cleaning the dataset, exploratory data analysis was conducted to understand customer behavior, identify churn patterns, and determine which factors were most strongly associated with customer churn. The EDA process includes descriptive statistics, visual analysis, correlation analysis, and statistical hypothesis testing.

A pie chart was created to examine the distribution of **customer status** categories. The dataset showed that more than one-quarter of customers had already churned, highlighting churn as a major business concern.

Descriptive statistics including mean, standard deviation, skewness, and kurtosis were calculated for numerical variable. Key findings included:

- Average customer age was 46.5 years.
- Average customer tenure was 32.4 months.
- Monthly charges averaged \$63.60.
- Total revenue averaged \$3034.38.
- Several financial variables such as *Total Refunds* and *Total Extra Data Charges* were heavily right-skewed with strong outliers.
- *Number of Dependents* and *Number of Referrals* were also positively skewed because most customers had very low values.

Mean comparisons by customer status showed that churned customers generally had shorter tenures, higher monthly charges and lower total revenue contribution than long-term retained customers.

Histograms were generated for all **numerical variables** to examine their distributions. The analysis revealed that:

- Age distribution was relatively balanced across adult age groups.
- Tenure was concentrated around very new customers and very long-term customers.
- Monthly charges showed a broad spread across pricing ranges.
- Total Charges and Total Revenue were strongly right-skewed, with many customers generating low revenue and a smaller group generating very high revenue.
- Refunds and extra data charges contained many zero values with a few extreme observations.

Histograms comparing customer groups were created for Age, Tenure in Months, Monthly Charges and Total Revenue. The plots showed that:

- Churned customers were concentrated at lower tenure ranges.
- Churned customers tended to pay higher monthly charges.
- Stayed customers generated substantially higher total revenue over time

Grouped percentage bar charts were created for **Contract Type, Internet Type, Offer and Payment Method**. Key findings included:

- Month-to-month contracts had the highest churn percentages, while two-year contracts had the lowest churn.
- Fiber optic customers showed higher churn rates than DSL or cable customers.
- Offer E had the highest churn percentage among promotional offers.
- Customers using mailed checks showed higher churn rates than customers using automatic payment methods or credit cards.

Service subscription rates for churned and stayed customers were compared across phone and internet services. The analysis showed that churned customers were less likely to subscribe to certain internet services.

Several **statistical tests** were conducted to determine whether observed differences were statistically significant. **T-tests** confirmed that churned customers paid significantly higher monthly charges and had significantly shorter tenures than retained customers.

ANOVA and Tukey tests showed that two-year contract customers paid significantly lower monthly charges than both month-to-month and one-year customers, and Fiber optic customers generated significantly higher revenue than DSL, cable, or customers with no internet service.

Chi-square tests were used to evaluate relationships between categorical variables and customer status. Contract Type, Internet Type, Offer, Payment Method, Married and Paperless Billing showed significant association.

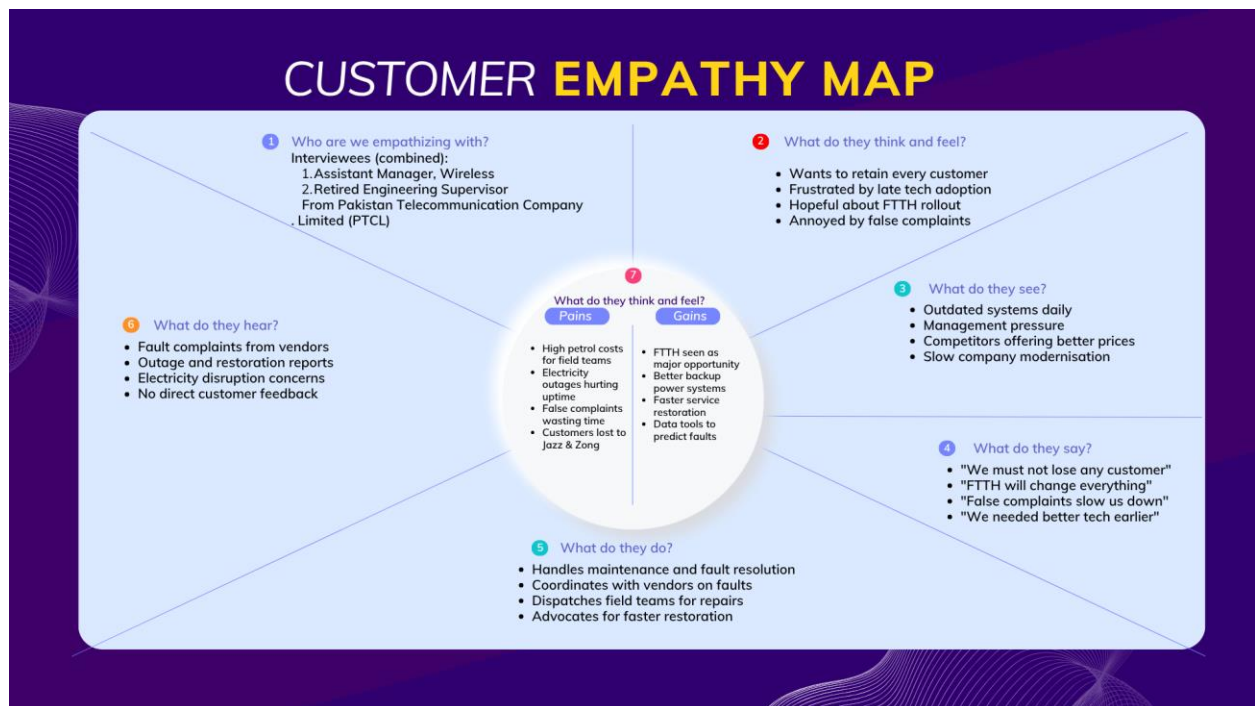
4. Design Thinking framework

4.1 Overview

The Design Thinking framework was applied across five stages: Empathize, Define, Ideate, Prototype, and Test, to structure the approach to solving the customer churn problem. Each stage built upon the previous one, ensuring that the final BI solution was grounded in real stakeholder needs and business context.

4.2 Empathize

The Empathize stage involved using AI as a domain expert proxy to simulate stakeholder interview questions. The team developed interview questions covering daily operations, decision-making challenges, customer lifecycle understanding, and data needs.



The empathy map captures the combined perspective of two PTCL professionals — an Assistant Manager in Wireless and a Retired Engineering Supervisor, summarising what they think, feel, see, hear, say, and do, along with their pains and gains.

In addition to the in class design sprint, the team conducted interviews with two professionals from Pakistan Telecommunication Company Limited (PTCL):

- **Mr. Aman** - Assistant Manager, Wireless, PTCL
- **Mr. Husnain** - Retired Engineering Supervisor, PTCL

The interview covered 17 questions spanning daily operations, customer churn causes, competitive landscape, retention strategies, and the role of technology in decision making.

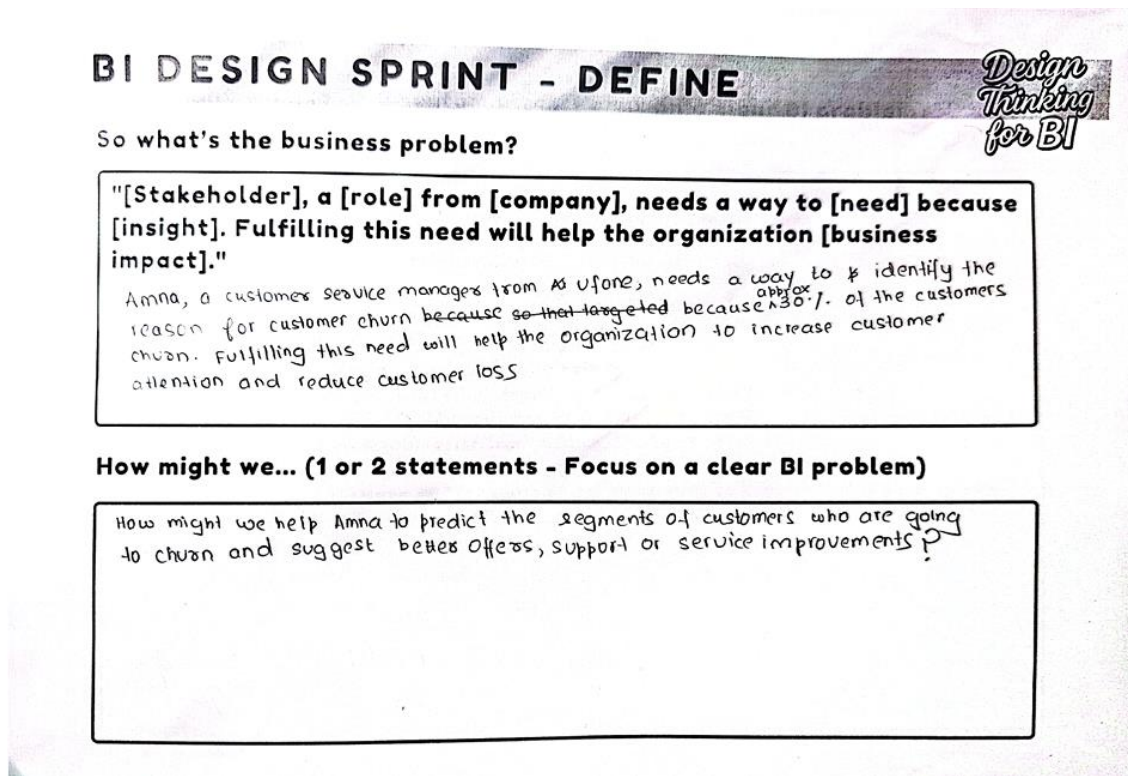
4.3 Insights

Key insights from the interview included:

- Both interviewees expressed a strong desire to retain every customer, viewing churn as a direct failure of the service
- PTCL faces significant pressure from competitors Jazz and Zong, who offer better pricing and faster network upgrades earlier (e.g. Fiber)
- The interviewees highlighted that late technology adoption was a major reason customers left PTCL in the past
- Operational challenges include high petrol costs for field maintenance, electricity outages affecting uptime, and time wasted on false complaints
- Mr. Aman was very optimistic about PTCL's Fiber to the Home (FTTH) rollout as a major opportunity to win back customers
- Decisions are largely task driven rather than data driven, with limited real time visibility into why customers leave
- The most common customer complaints were related to service quality and delays in resolution

These insights reinforced findings from the dataset, particularly the importance of service quality, competitive pricing, and proactive retention strategies.

4.4 Define



The Define stage focused the team's problem around a specific stakeholder need, producing a clear problem statement and How Might We question.

A hypothetical stakeholder persona was created to anchor the problem statement:

Problem Statement: "Amna, a customer service manager, needs a way to identify the approximately 30% of customers at risk of churning so that targeted interventions can be applied. Fulfilling this need will help the organisation increase customer retention and reduce customer loss."

How Might We Statement: "How might we help Amna predict the segments of customers going to churn and suggest better offers, support, or service improvements?"

4.5 Ideate

BI DESIGN SPRINT - IDEATE

Design Thinking for BI

TEAM MEMBER 1 NAME

Come up with as many ideas. One idea per sticky note.

a chart showing demographic vs churn	Total charges when churned	churn reason with churned	online security with churn	Internet type analysis
Area (long, lat) with churn	Tenure Analysis new vs old	Customer Experience (referral etc)	high risk	Contract type (long term)
Revenue at Risk	Service bundle	Referral effect	Risk Scoring	

No judgement - Go for volume - Be creative

generated ideas including demographic vs churn charts, revenue at risk analysis, tenure analysis, service bundle analysis, and referral effect visualisations.

BI DESIGN SPRINT - IDEATE

*Design
Thinking
for BI*

TEAM MEMBER 2 NAME Hadiya Muneeb

Come up with as many ideas. One idea per sticky note.



No judgement - Go for volume - Be creative

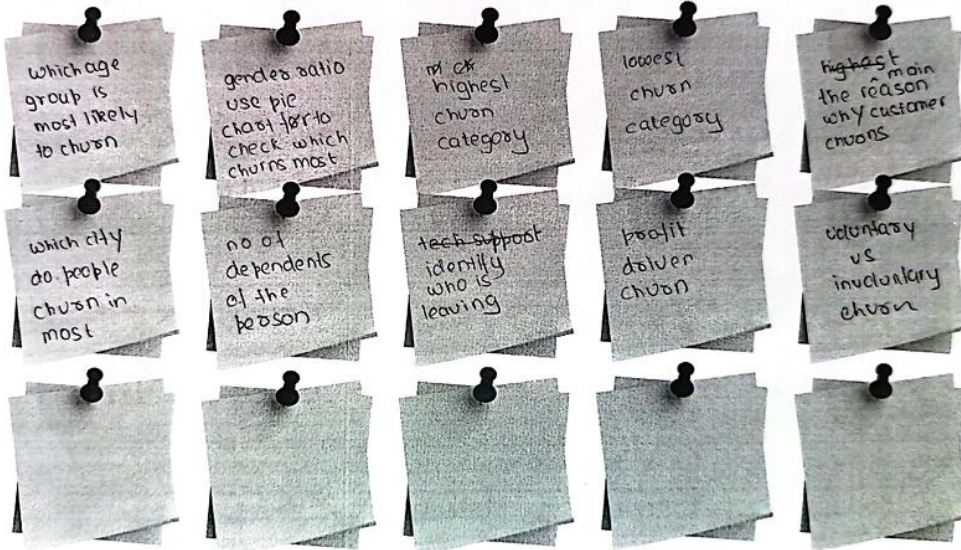
contributed ideas including age group identification, churn trend by month, churn by city and region, monthly charges vs churn, and referrals vs churn analysis.

BI DESIGN SPRINT - IDEATE

*Design
Thinking
for BI*

TEAM MEMBER 3 NAME

Come up with as many ideas. One idea per sticky note.

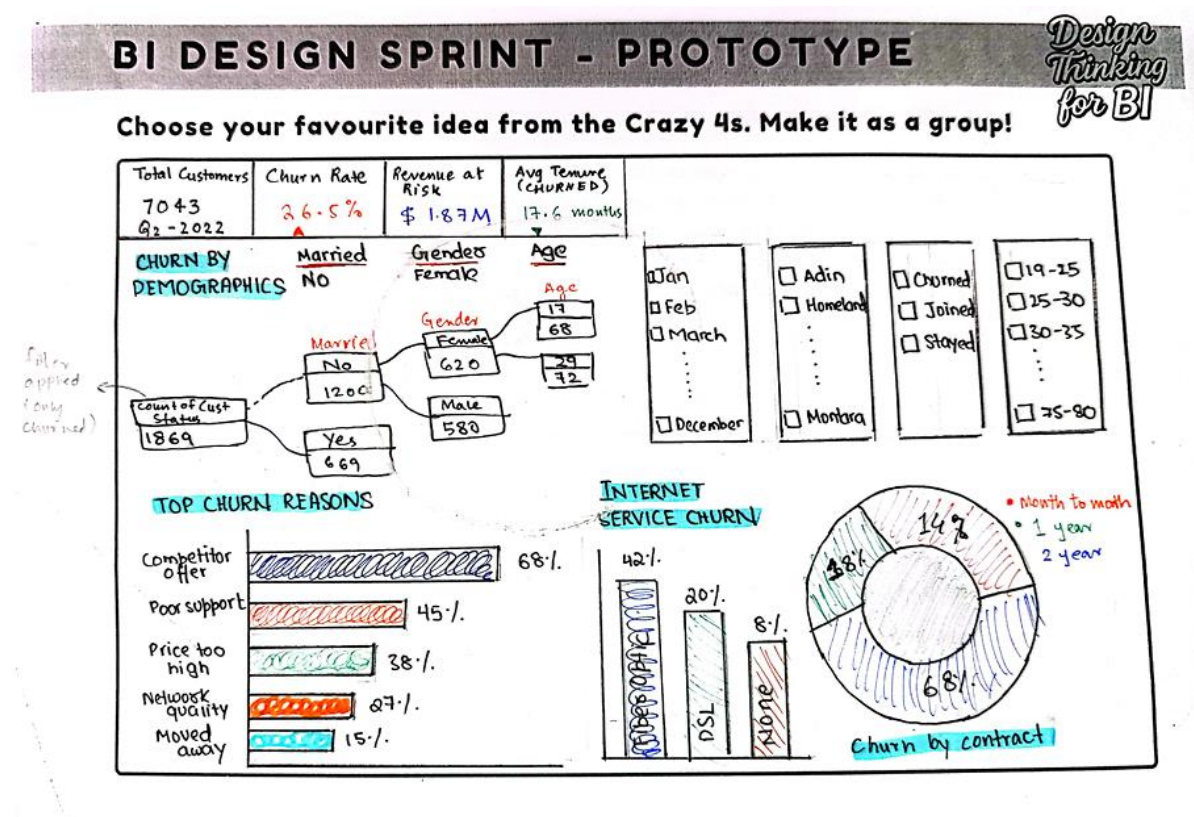


No judgement - Go for volume - Be creative

ideated on age group churn rates, gender ratio analysis, highest and lowest churn categories, tech support identification, and voluntary vs involuntary churn.

Each team member individually generated ideas through the Crazy 4s exercise. The ideas that were ultimately implemented in the final Power BI report include demographic analysis, revenue and charges analysis, service subscription analysis, churn reasons analysis, and contract type comparisons.

4.6 Prototype



The group prototype combined the strongest individual ideas into a single dashboard sketch covering KPI cards (Total Customers, Churn Rate, Revenue at Risk, Avg Tenure), churn by demographics, top churn reasons, internet service churn, and churn by contract type with filters for Customer Status, City, Month and Age Group.

The prototype was sketched as a group and served as the blueprint for the final Power BI solution. It included:

- KPI cards showing Total Customers (7,043), Churn Rate (26.5%), Revenue at Risk (\$1.87M), and Average Tenure of Churned Customers (17.6 months)
- Churn by Demographics section with filters for gender, age, and marital status
- Top Churn Reasons ranked by frequency
- Internet Service Churn breakdown by type
- Churn by Contract Type showing Month to Month, One Year, and Two Year

4.7 Test

BI DESIGN SPRINT - TEST

*Design
Thinking
for BI*

Upload the prototype picture & ask your AI expert for feedback!

Our AI expert says this is a strong prototype with clear KPIs and useful churn insights, especially revenue at risk and top reasons. It helps the expert understand what's happening, but not what actions to take next. It would need clear priorities and trend data ~~to track data~~ to track changes over time. The demographics section could be simpler.

What can you improve? Reflect as a team!

It's missing a time trend line to see if churn is getting worse week over week not just a Q2 timesame. ~~xx~~ We could also add a map to highlight the ^{for} problematic areas based on zip code. Also, an average monthly charge for churned vs retained because price sensitivity is invisible right now

The Test stage involved submitting the prototype to an AI expert for feedback, followed by a team reflection on what could be improved.

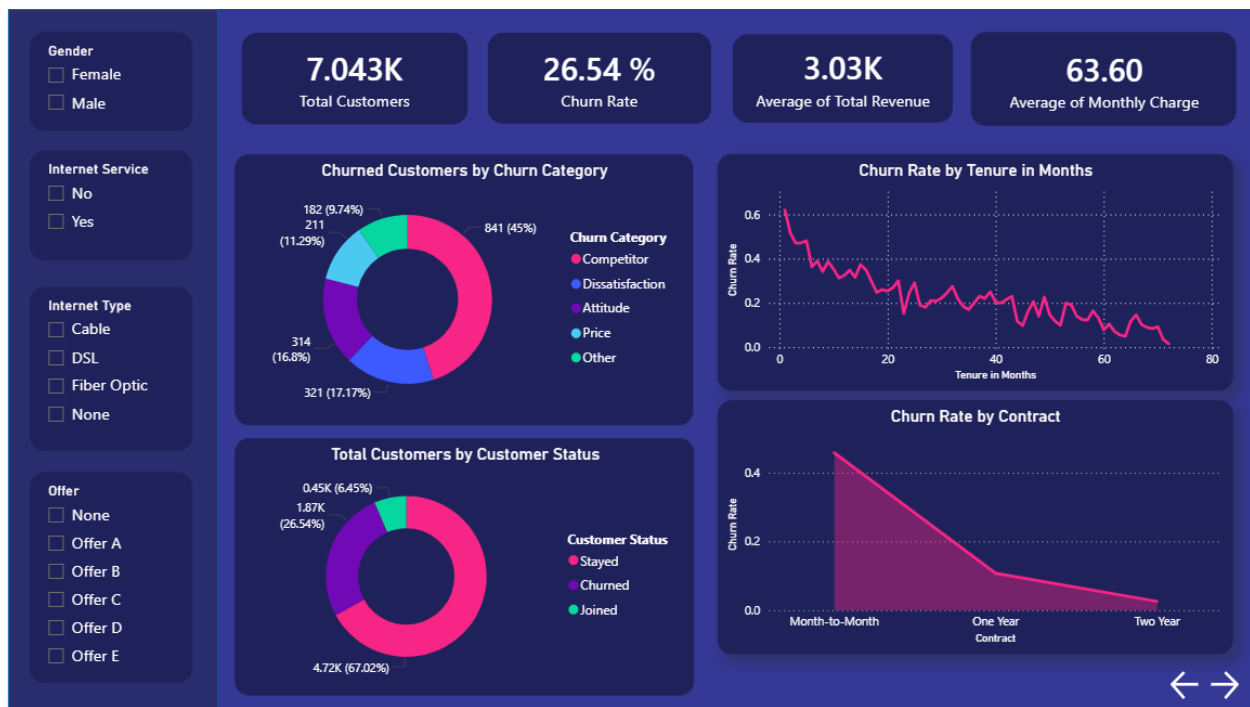
The AI expert feedback on the prototype noted that it was a strong design with clear KPIs and useful churn insights, particularly around revenue at risk and top churn reasons. It helped the stakeholder understand what was happening but lacked clear priorities and trend data to show change over time. Specific improvement suggestions included adding a time trend line, a geographic map visual, and simplifying the demographics section.

Team Reflection: We identified three areas for improvement: adding a time trend to track churn over the customer lifecycle, a map to highlight problematic geographic areas, and an average monthly charge comparison between churned and retained customers to make price sensitivity visible.

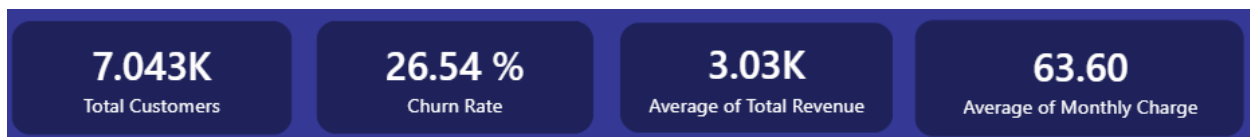
These reflections directly shaped the final dashboard design. The churn rate by tenure line chart addresses the time trend feedback. The churn by city treemap addresses the geographic feedback. The average monthly charge comparison by customer status directly responds to the pricing visibility gap identified during testing.

5. BI Charts

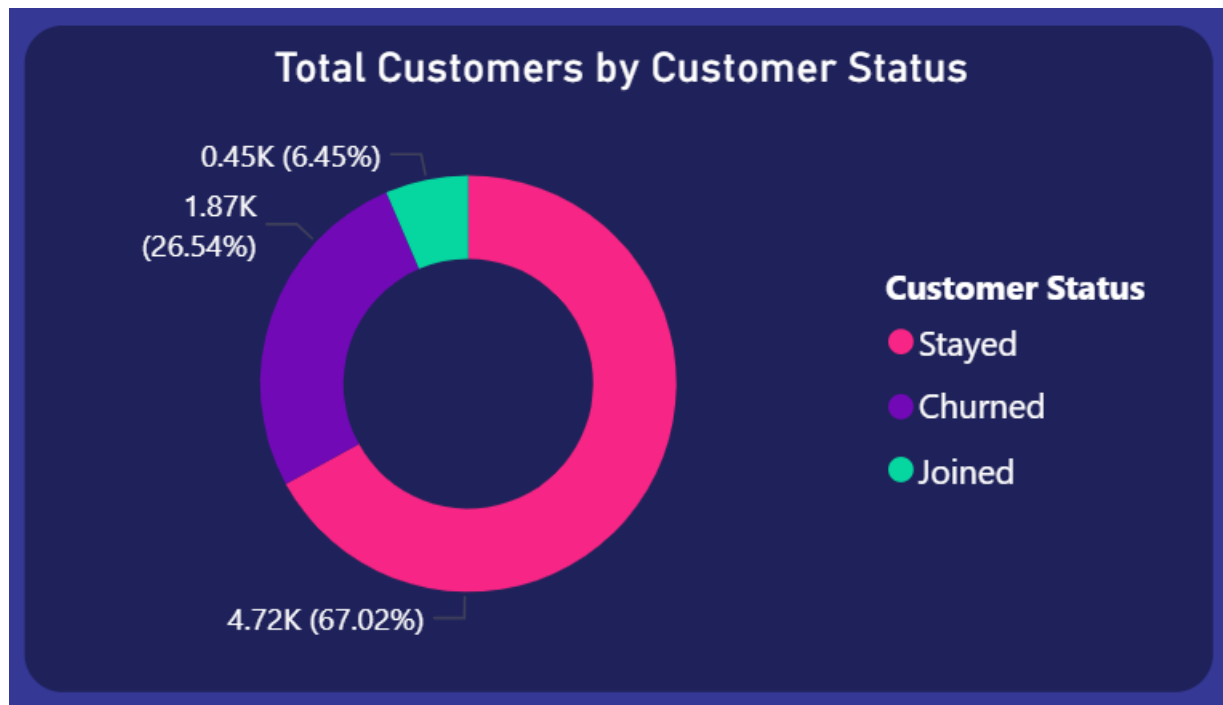
5.1 Overview



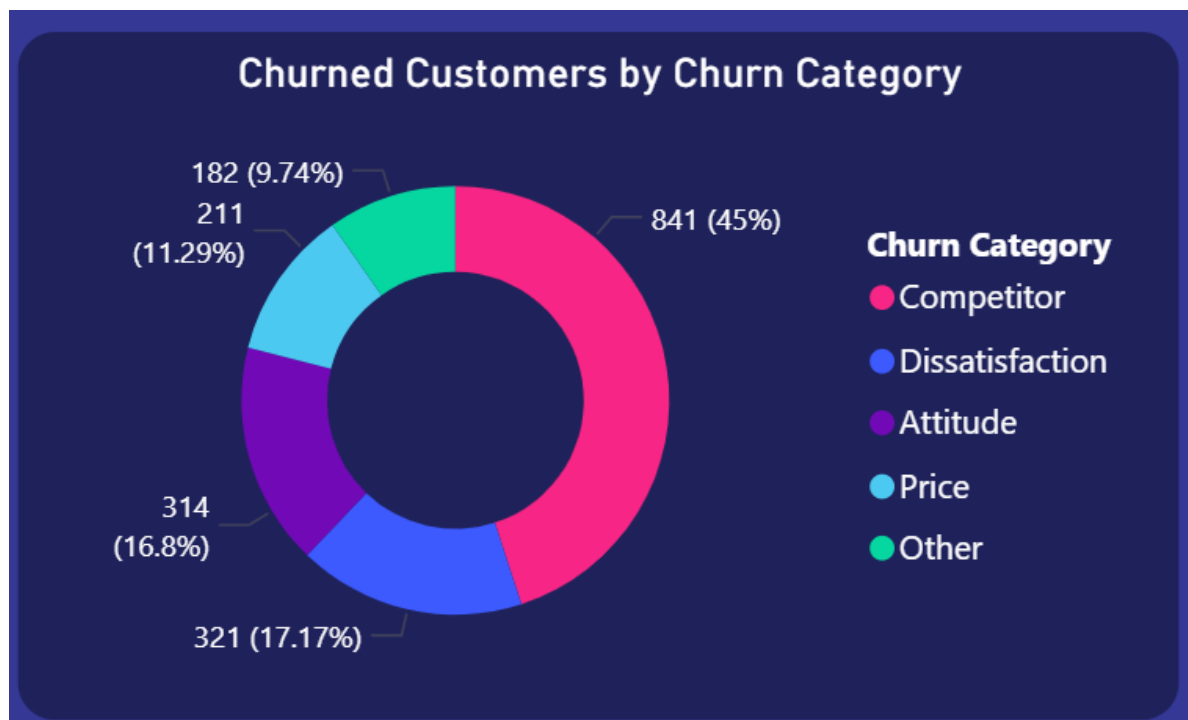
This dashboard highlights the overall churn patterns and shows that customer loss is strongly tied to customer retention strategies and contract structure. Churn is concentrated among customers with shorter tenures and month-to-month contracts, suggesting that customers who are not locked into long-term commitments are far more likely to leave. The top Churn reasons imply that customers are highly sensitive to competing offers, pricing, and service quality in the telecom market. The dashboard also suggests that customer loyalty increases over time, as churn steadily declines for long-tenure customers.



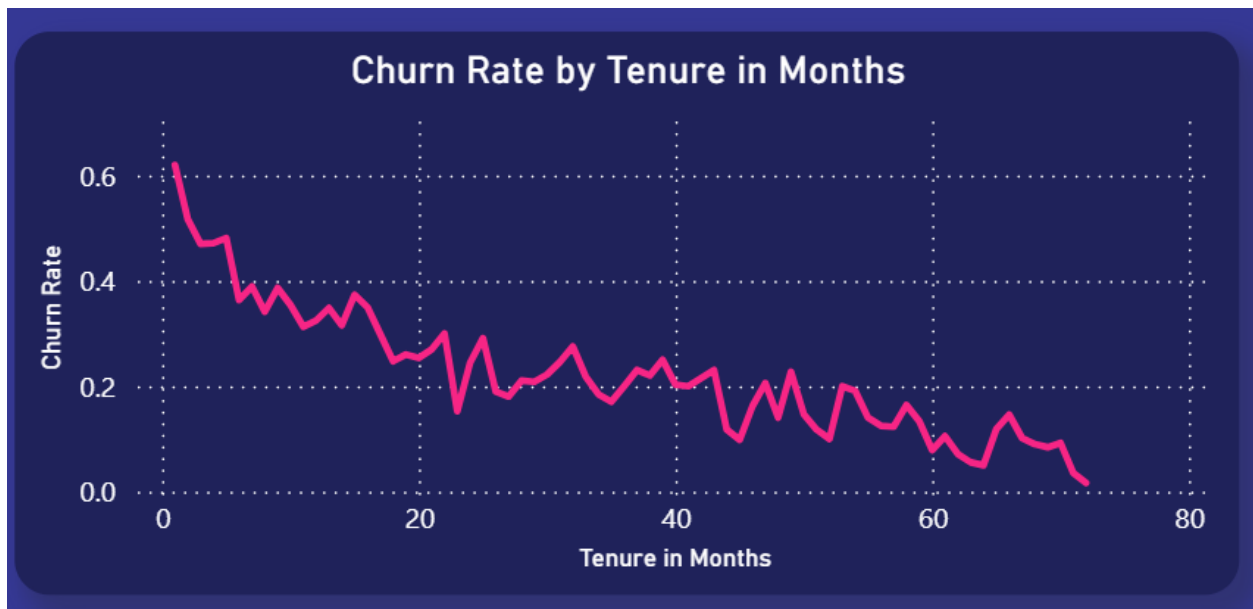
The overall churn rate is 26.54%, indicating that more than one-quarter of customers have left the company. The average revenue generated per customer is approximately 3.03K, showing the financial contribution of the customer base. Customers pay an average monthly bill of 63.60, providing a baseline for pricing and billing analysis.



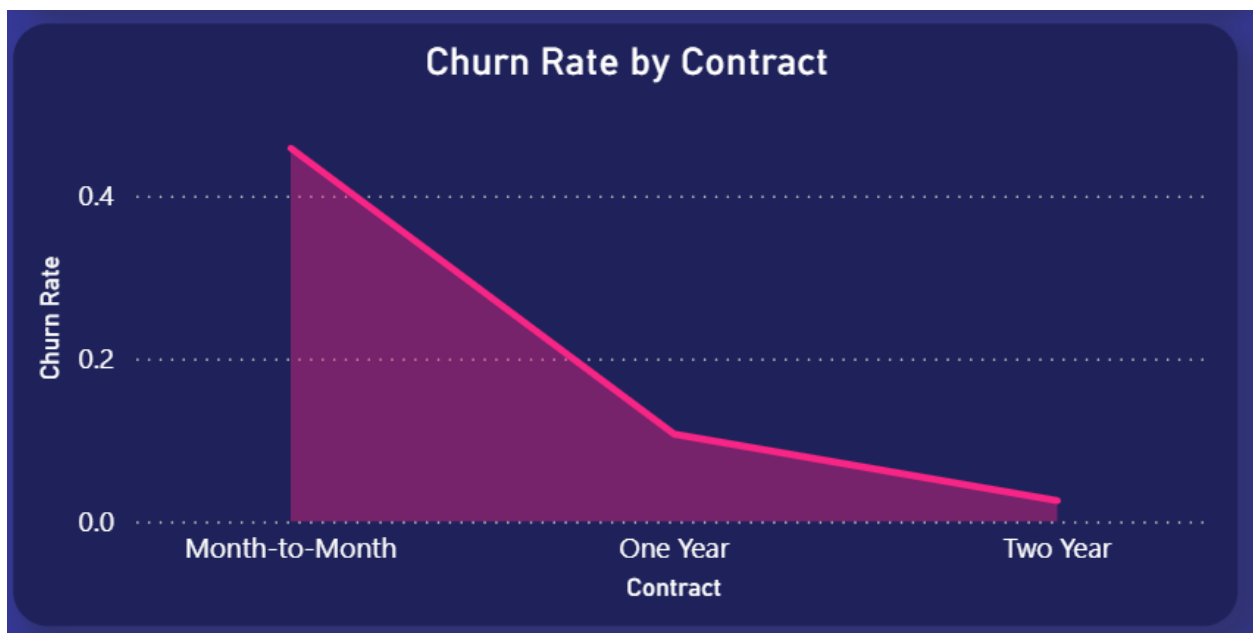
Most customers remained with the company, while churned customers formed roughly one-quarter of the customer base and newly joined customers represented the smallest segment.



Competitor-related reasons account for the largest portion of churned customers, followed by dissatisfaction and attitude-related issues. Price-related reasons contribute the least, albeit still a significant amount.

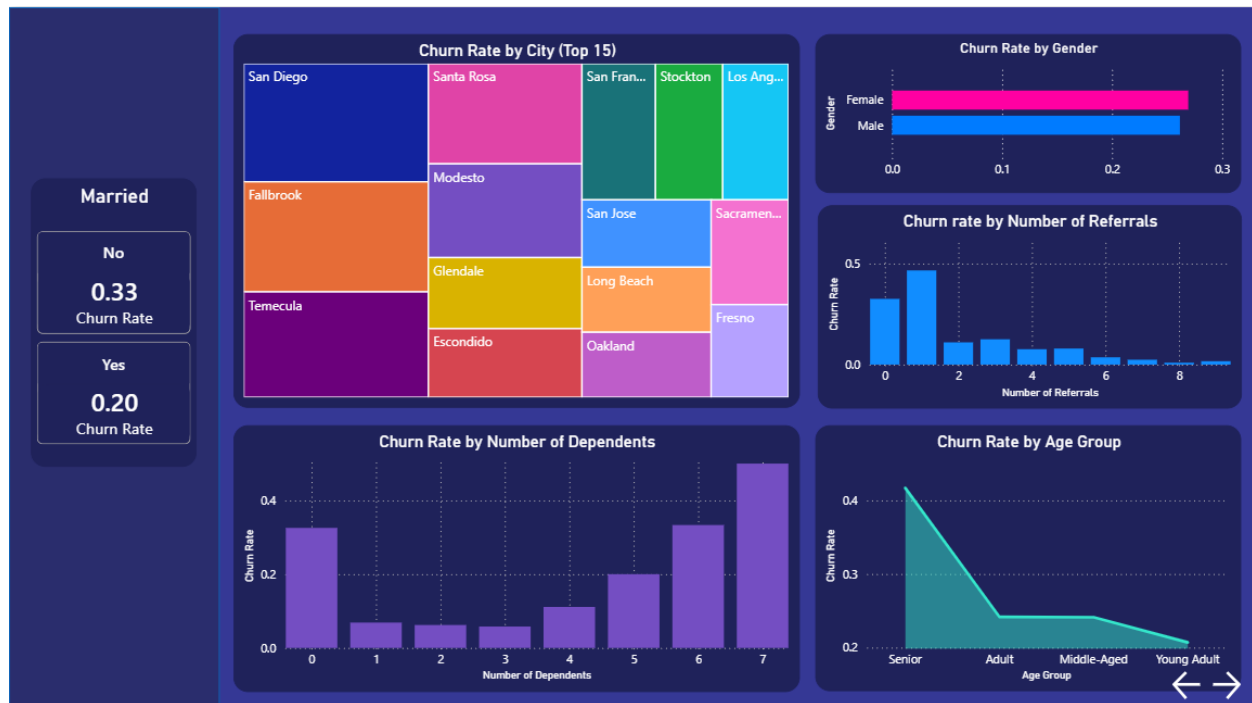


Churn rate is highest during the early months of customer tenure and gradually decreases as customers stay longer with the company. Long-term customers are less likely to churn.

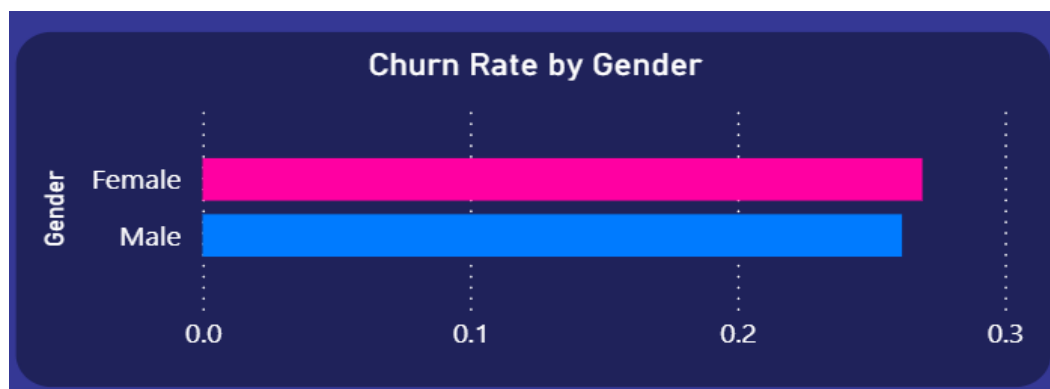


Month-to-month contracts have the highest churn rate, while one-year and two-year contracts show significantly lower churn. Longer contracts are associated with better customer retention.

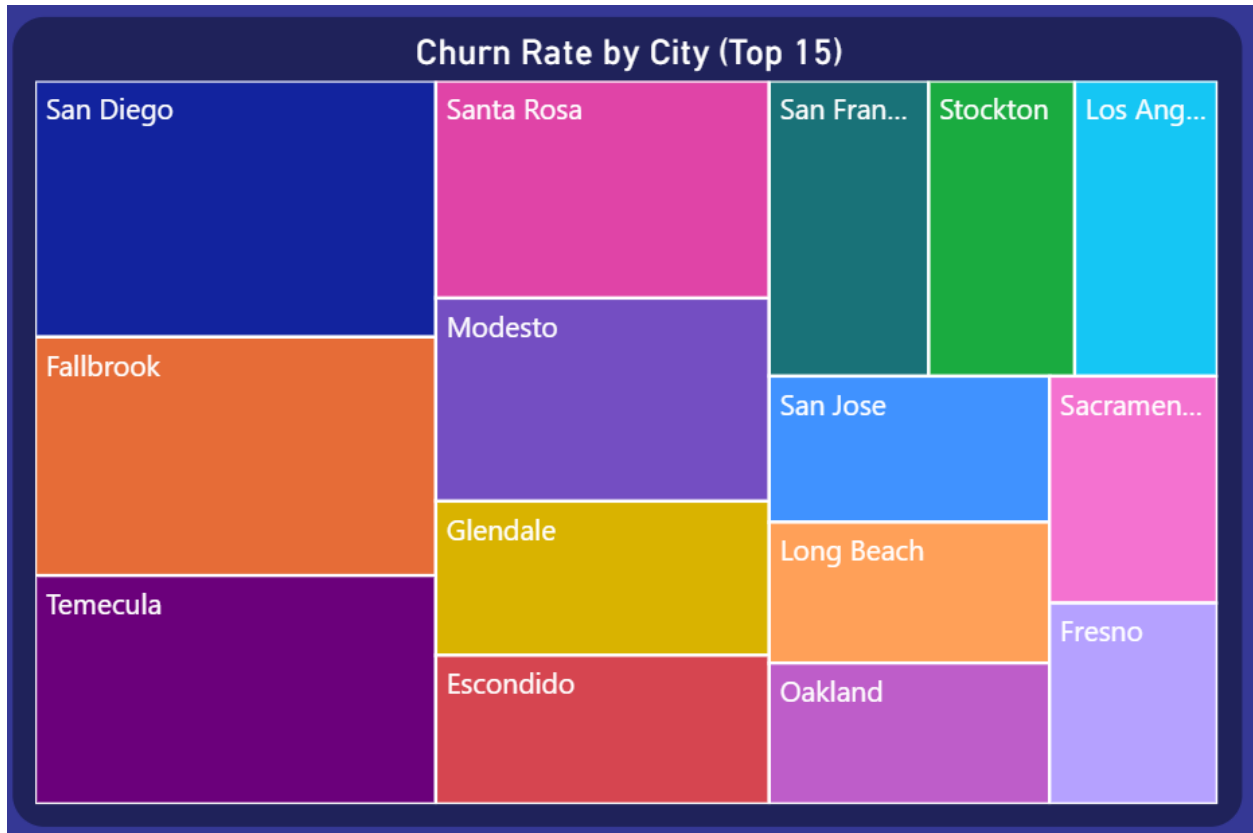
5.2 Customer Demographic Analysis



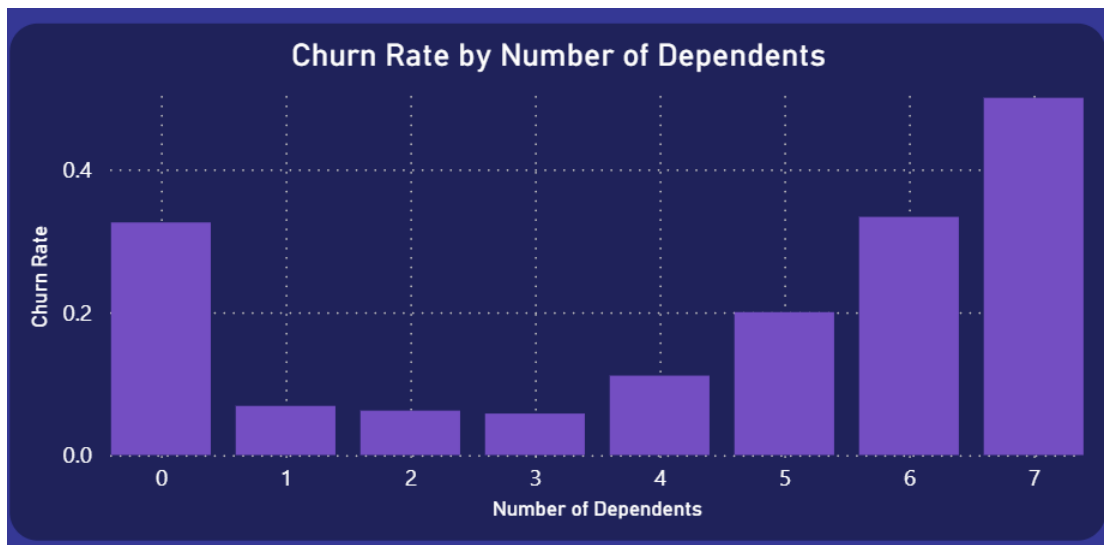
This dashboard focuses on demographic and geographic factors influencing churn. Customers with no dependents, fewer referrals, and unmarried status experience noticeably higher churn rates, indicating that these customers may have weaker attachment to the service provider and are more high risk. Senior customers also display the highest churn rates, which may point toward dissatisfaction with pricing, support quality, or service usability. Geographic variation across cities suggests that regional competition and service quality differences may affect retention.



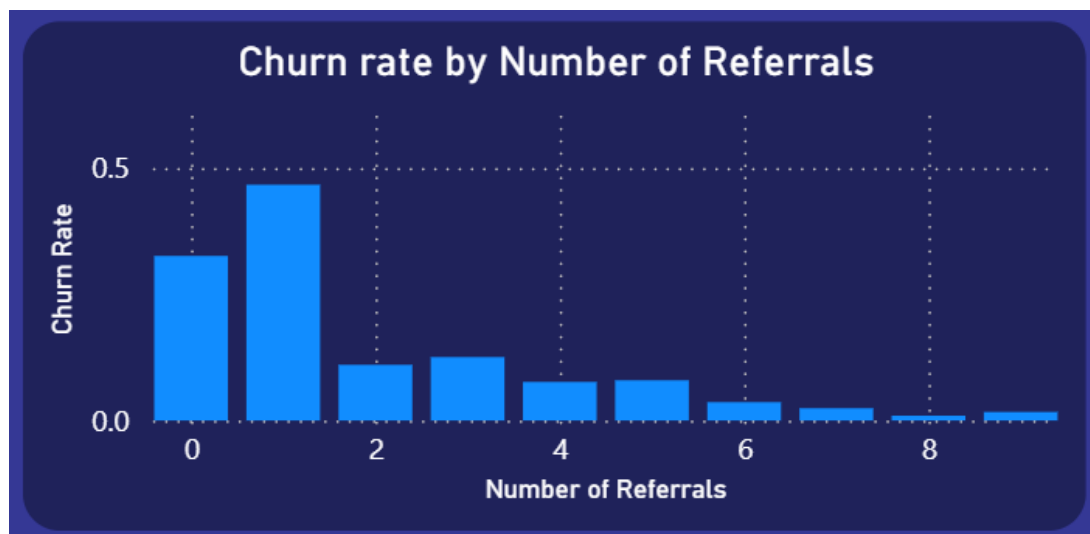
Female customers show a slightly higher churn rate than male customers, though the difference is insignificant. This was also revealed by the statistical tests.



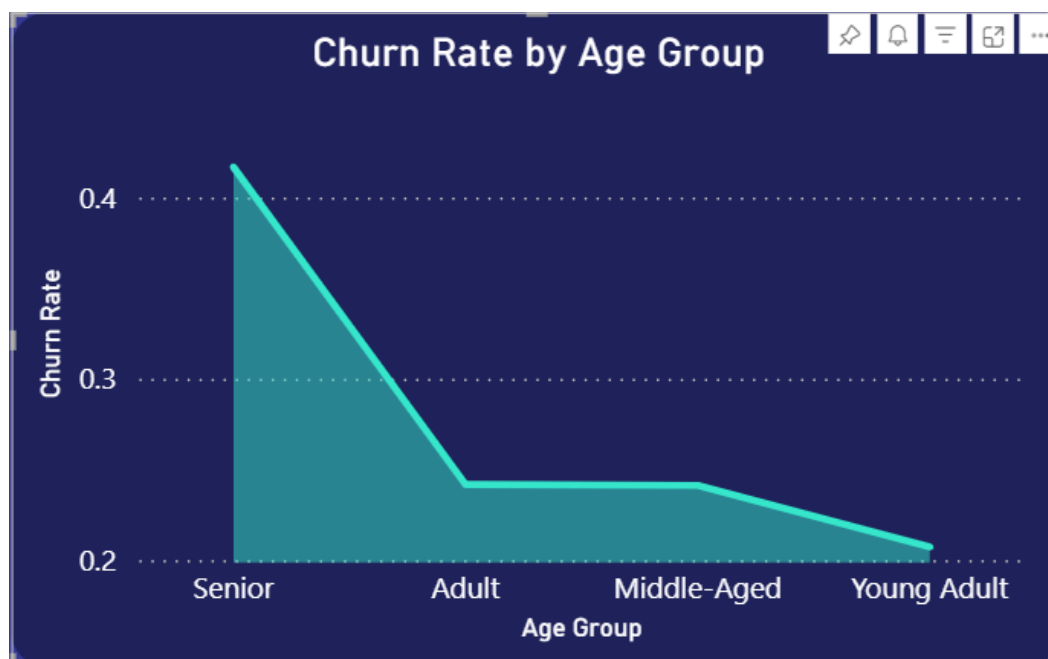
Churn rates vary considerably across cities, with some cities contributing more heavily to customer loss. Geographic location appears to influence churn behavior. These locations will be investigated further to pinpoint the cause of higher churn rates.



Customers with no dependents or higher than 4 dependents experience higher churn rates, while customers with few dependents tend to remain more loyal.



Customers with more referrals generally have lower churn rates, indicating that engaged and loyal customers are more likely to refer others.



Senior customers have the highest churn rate among all age groups, while young adults show the lowest churn tendency.

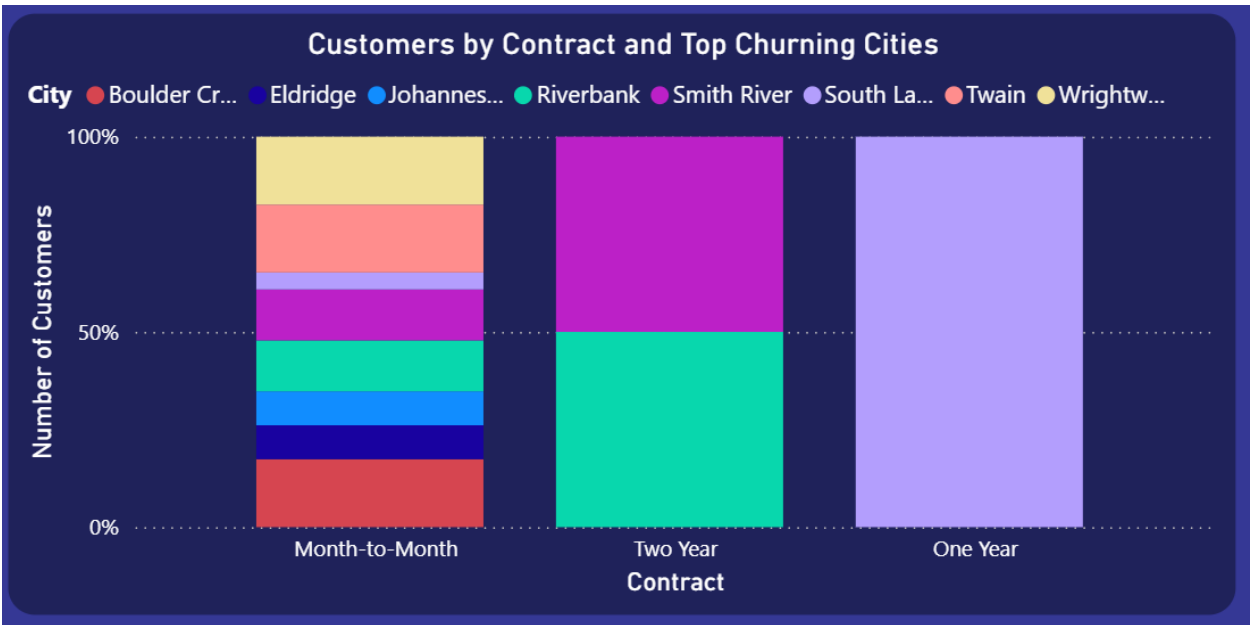


Unmarried customers have a noticeably higher churn rate compared to married customers, suggesting stronger retention among married households.

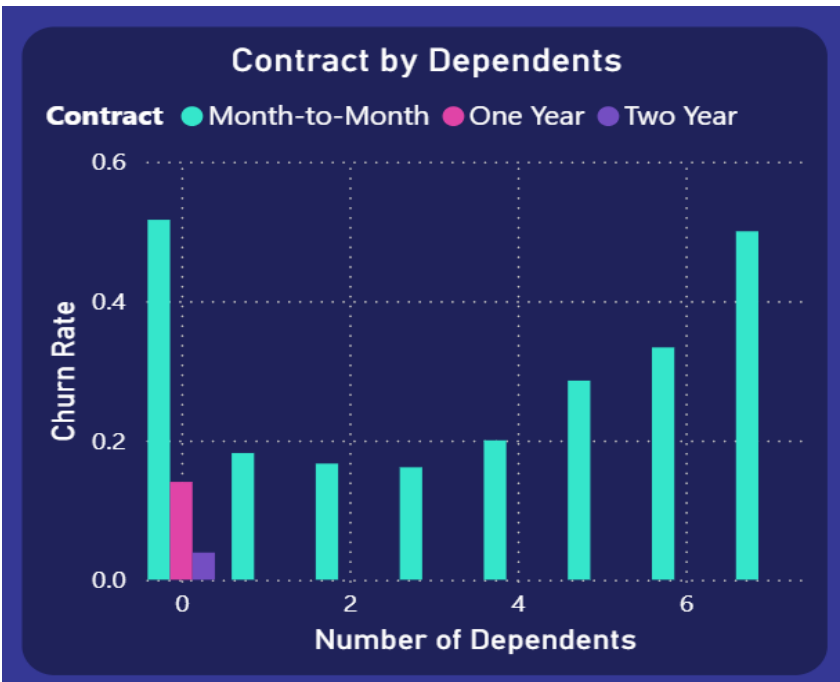


This dashboard further investigates churn patterns to identify why certain customer groups and locations experience higher churn rates. Month-to-month contracts are consistently associated with high churning customer segments and cities, while long-term contracts are noticeably less common in top churning areas, suggesting they improve retention. Customers with more referrals also tend to hold longer contracts, indicating that satisfied customers are both more loyal and more likely to recommend the company. The dashboard additionally shows that churn prone groups are less likely to subscribe to premium tech support, implying that support services may help reduce churn. Variations

in monthly charges across offers also suggest that pricing and promotional strategies influence customer retention behavior.

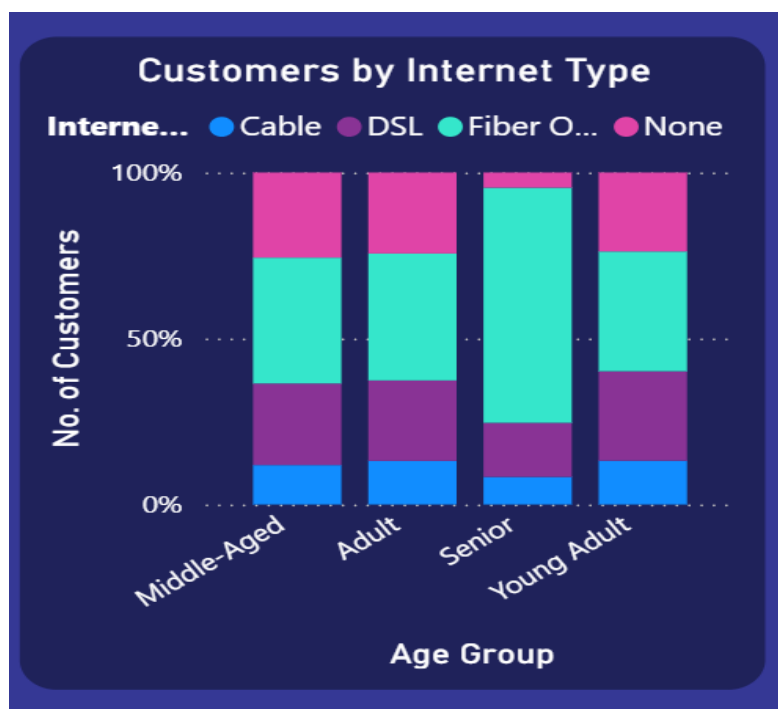


Top churning cities have a much higher proportion of Month-to-month contracts, while long-term contracts appear rarely.

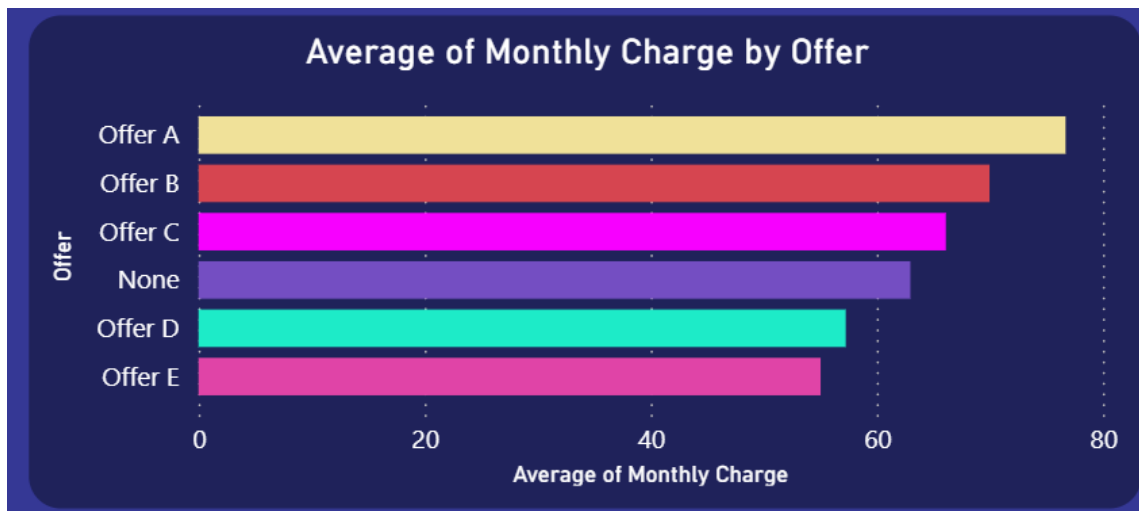


Due to limited data for customers with more than 3 dependents, the metrics for those cannot be considered. A greater proportion of customers with 1-3 dependents prefer longer contracts compared to customers with 0 dependents.

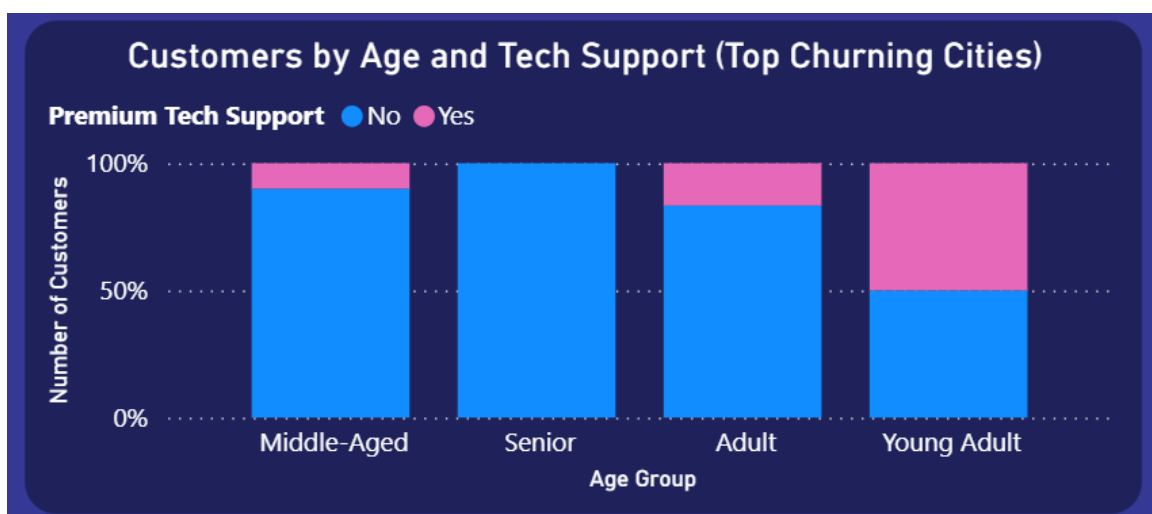
Contract	Month-to-Month		One Year		Two Year		Total	
Number of Dependents	Churn Rate	No. of Customers	Churn Rate	No. of Customers	Churn Rate	No. of Customers	Churn Rate	No. of Customers
0	0.52	2999	0.14	1183	0.04	1234	0.33	5416
1	0.18	209		115		229	0.07	553
2	0.17	198		127		206	0.06	531
3	0.16	186		123		208	0.06	517
4	0.20	5				4	0.11	9
5	0.29	7		2		1	0.20	10
6	0.33	3					0.33	3
7	0.50	2					0.50	2
8						1		1
9		1						1
total	0.46	3610	0.11	1550	0.03	1883	0.27	7043



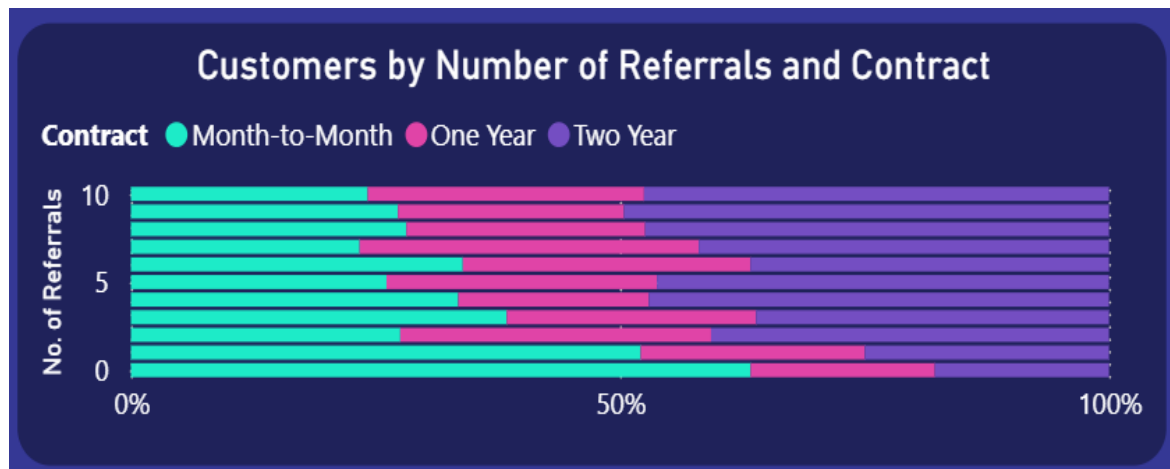
Fiber Optic internet is widely used across age groups, while customers without internet service form a smaller proportion of the customer base.



Offer A customers pay the highest average monthly charges, while Offer E customers pay the lowest. Different marketing offers are associated with different customer spending patterns.

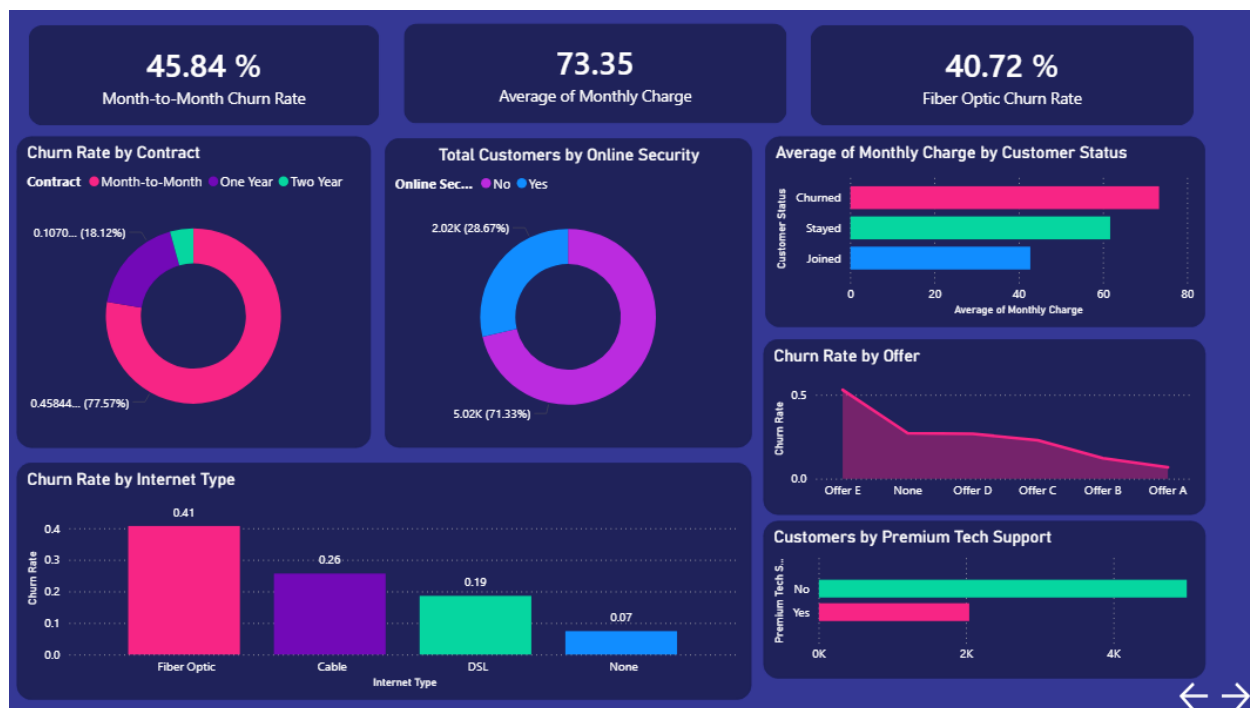


Customers without premium tech support are more concentrated among churn-prone groups, particularly Seniors, Adults and Middle-aged individuals.

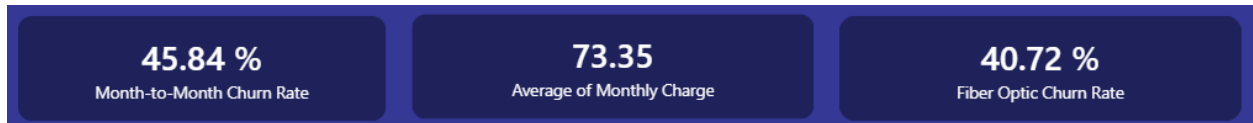


Customers with more referrals are more commonly associated with longer-term contracts, suggesting stronger customer loyalty and engagement.

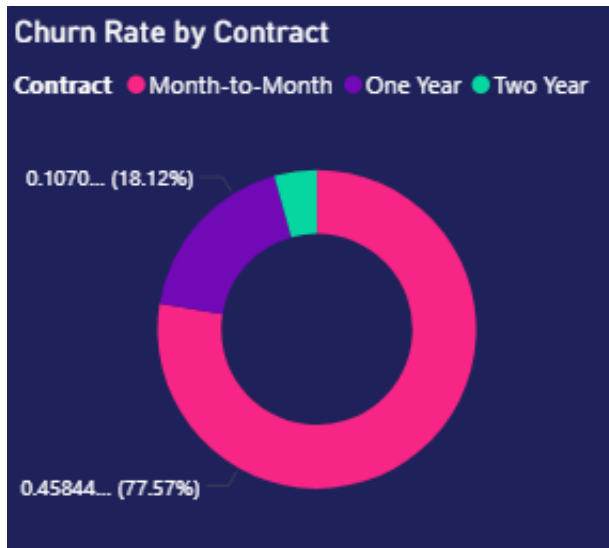
5.3 Services and Subscription



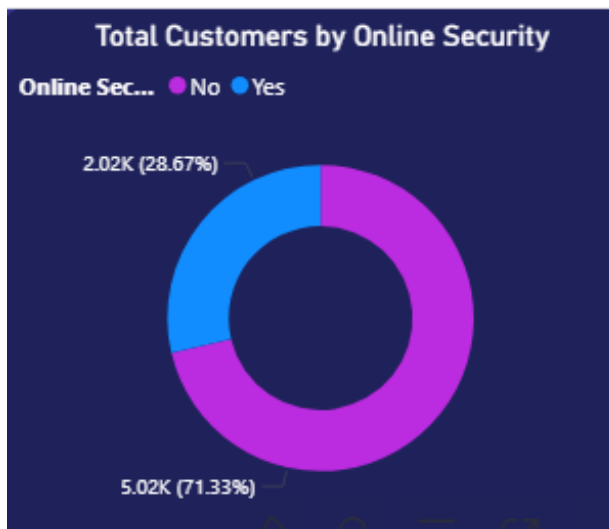
This dashboard shows that service subscription choices and contract type are the strongest predictors of churn. Month-to-month customers are more likely to leave, and fiber optic customers churn the most despite paying the highest charges, and customers without add-on services show higher churn rates across all categories.



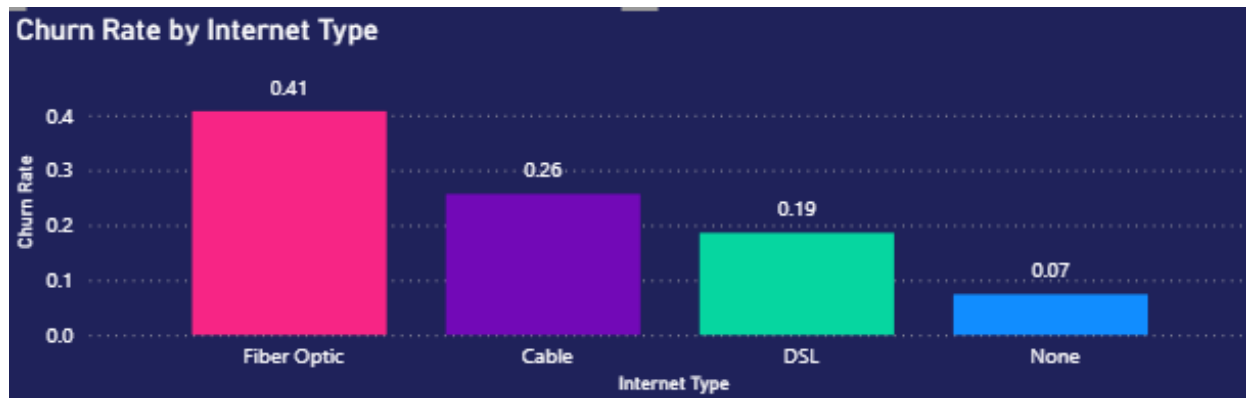
Month-to-month contract customers churn at 45.84%, nearly fifteen times higher than two-year contract customers. Churned customers pay an average of \$73.35 per month while fiber optic customers churn at 40.72% despite being on the most premium internet category.



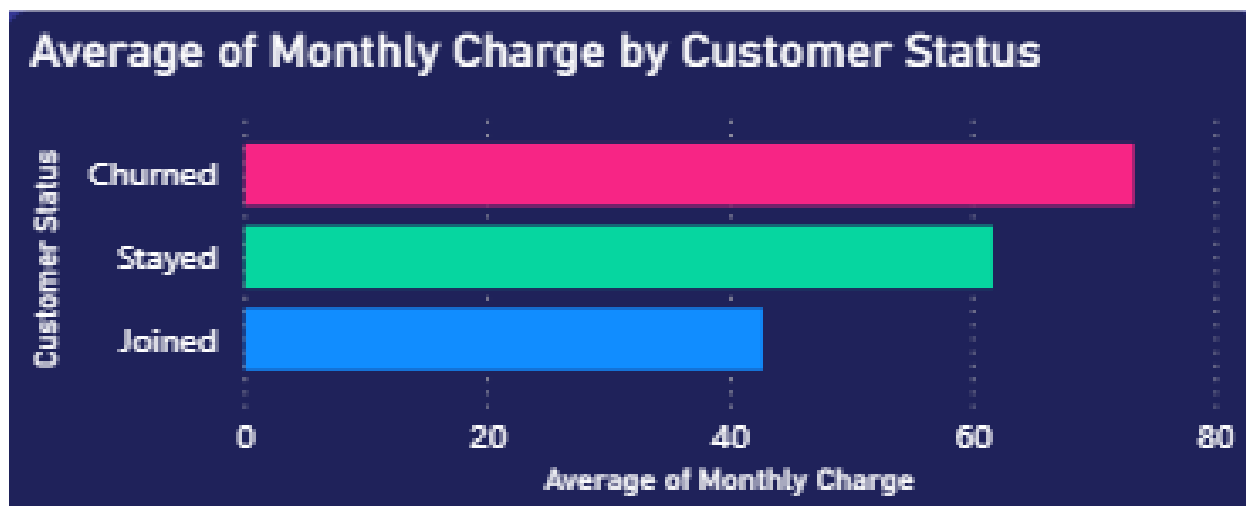
Month-to-month customers consist of 77.57% of all churned customers while two-year contract customers account for just 18.12%, showing that contract length is determinant of customer loyalty.



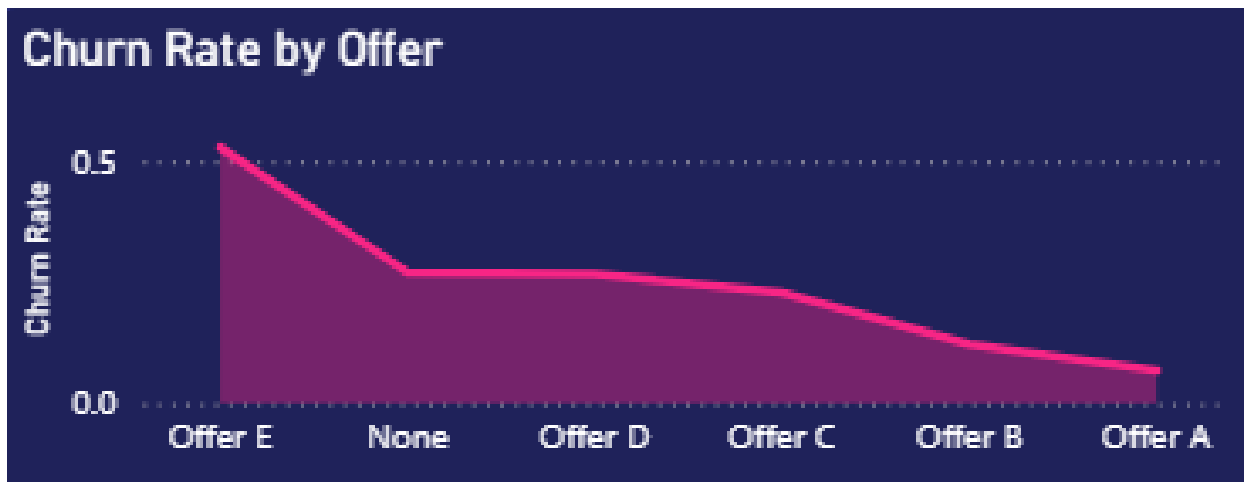
71.33% of customers do not use online security services, showing low adoption of add-ons. Customers without these services are more likely to churn



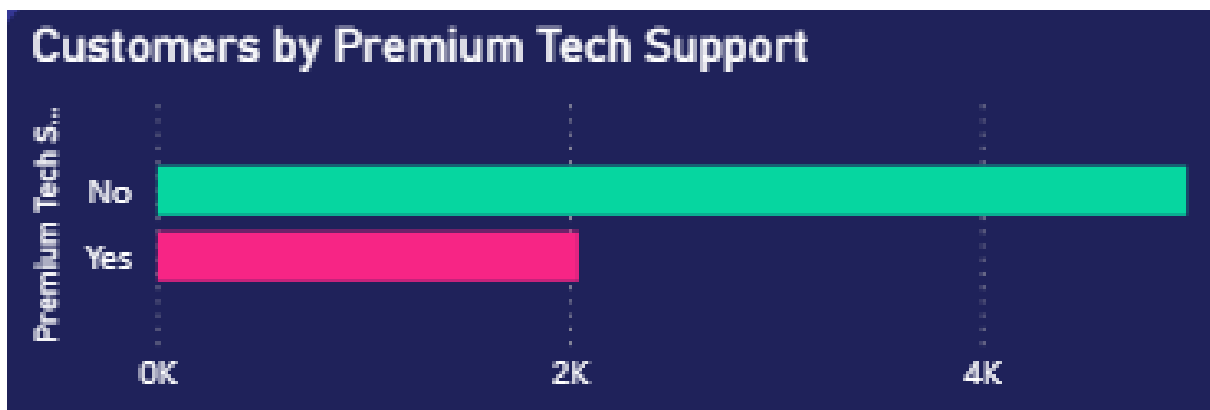
Fiber optic users have the highest churn rate at 41%, compared to 26% for cable and 19% for DSL users. This suggests that problems with customer satisfaction are mostly in fiber type.



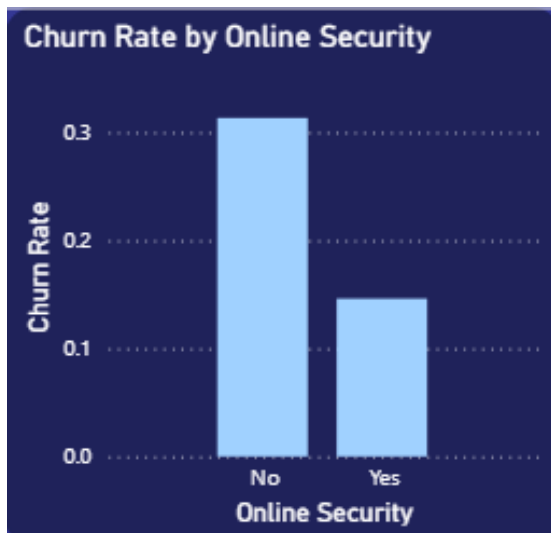
Churned customers have the highest average monthly charges at \$73.35, compared to \$61.74 for retained customers and \$42.78 for new customers. This means the company is losing many of its high-paying customers.



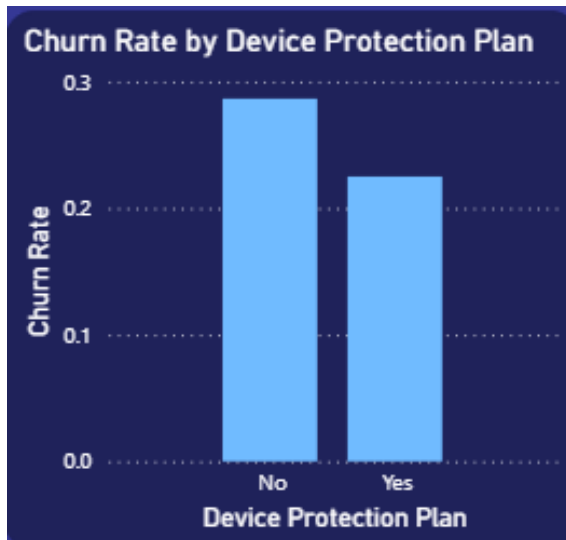
Offer E has the highest churn rate at around 52%, making it the least effective offer, while Offer A has the lowest churn rate and appears to be the best for customer retention.



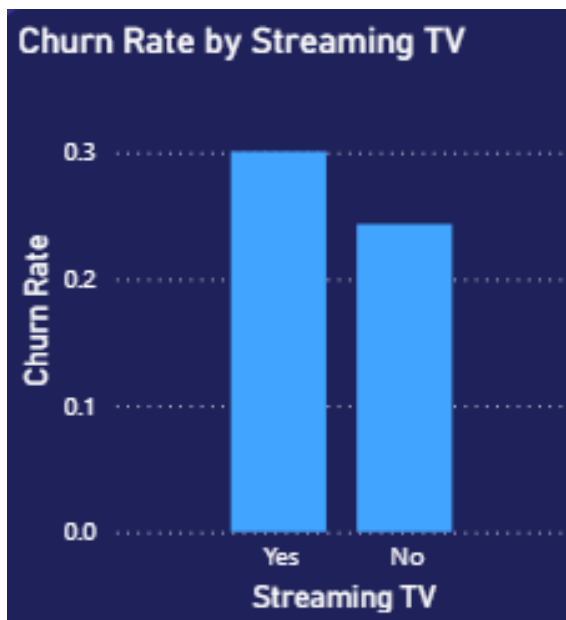
Most customers do not subscribe to premium tech support, and this group shows higher churn levels. Customers with tech support seem more likely to stay loyal.



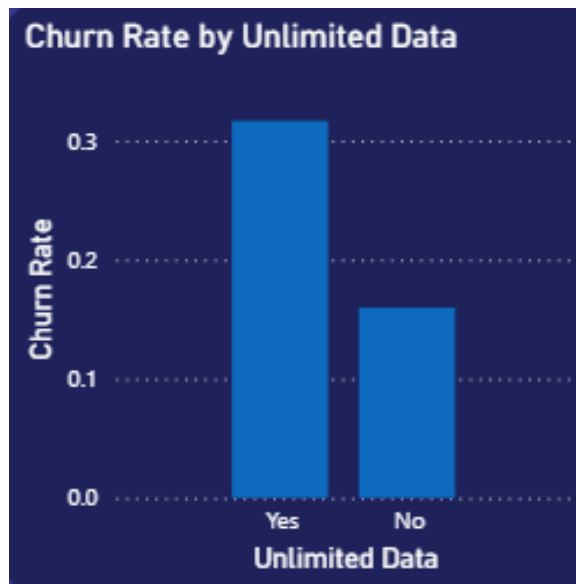
Customers without online security have a churn rate of 32%, compared to 15% for customers with it. This shows that online security strongly helps retain customers.



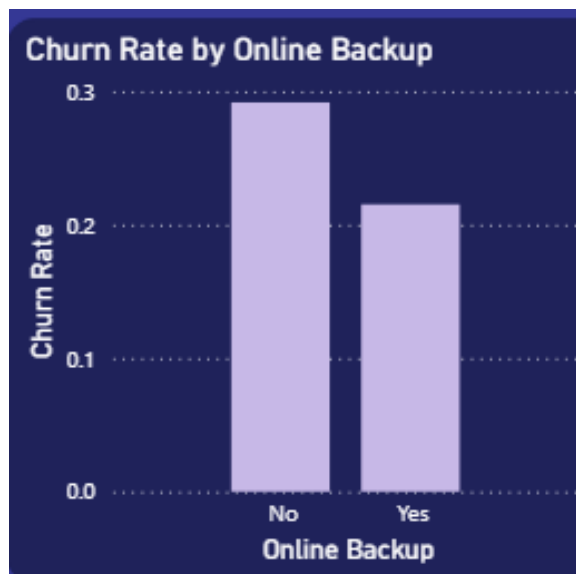
Customers without a device protection plan churn about 29%, while those with the plan churn at 23%. Protection plans slightly reduce churn by increasing customer dependence on the service.



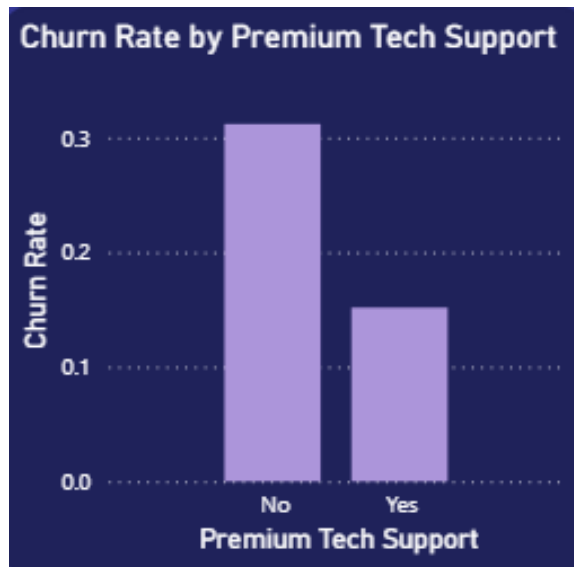
Customers without streaming TV churn at 30%, compared to 24% for customers who subscribe to it. Entertainment services appear to improve customer retention.



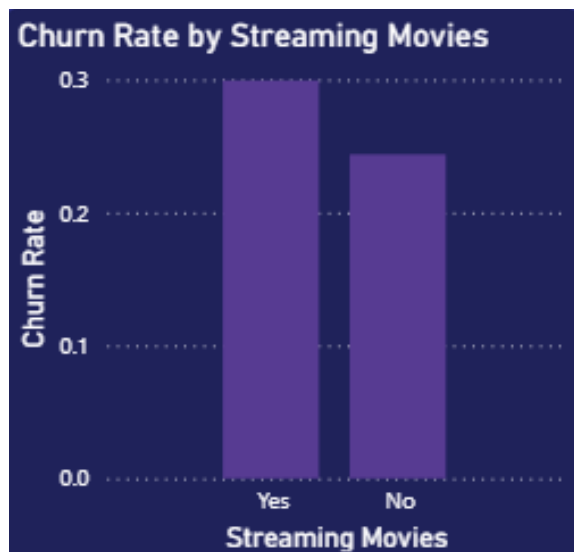
Customers with unlimited data churn at 31%, much higher than the 16% churn rate for customers without it. This suggests unlimited data users may be more price-sensitive or have higher service expectations.



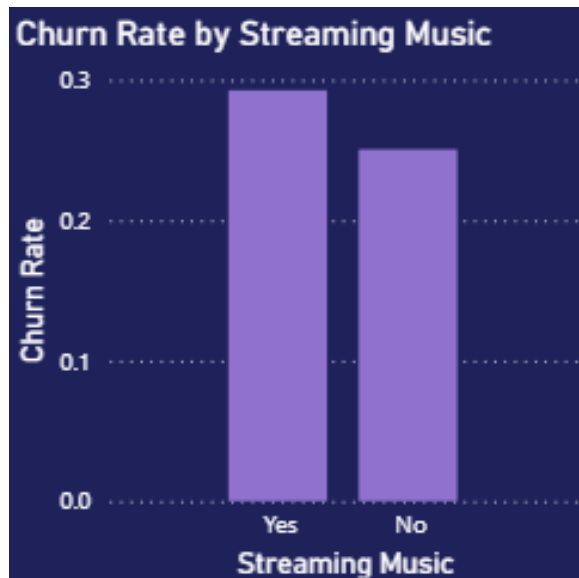
Customers without online backup churn at 29%, compared to 22% for those with it. Online backup services help improve customer loyalty.



Customers without premium tech support churn at 31%, while those with it churn at only 15%. Premium tech support is one of the strongest services for reducing churn.

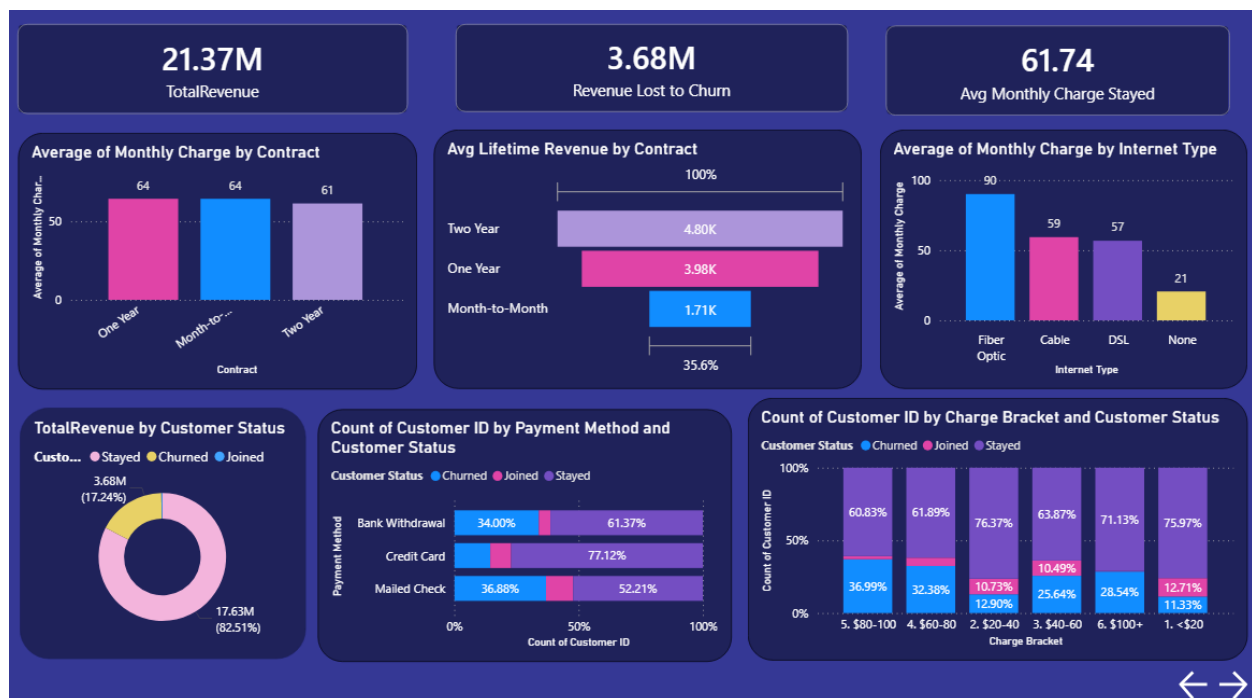


Customers with streaming movies churn at 30%, while those without churn at 24%. Streaming movies has minimal impact on retention decisions.

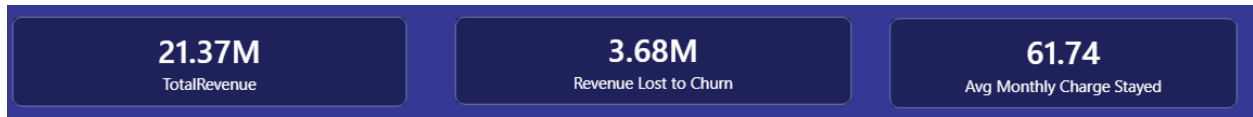


Customers with streaming music churn at 29%, compared to 25% for those without it. Streaming music has very little effect on customer retention.

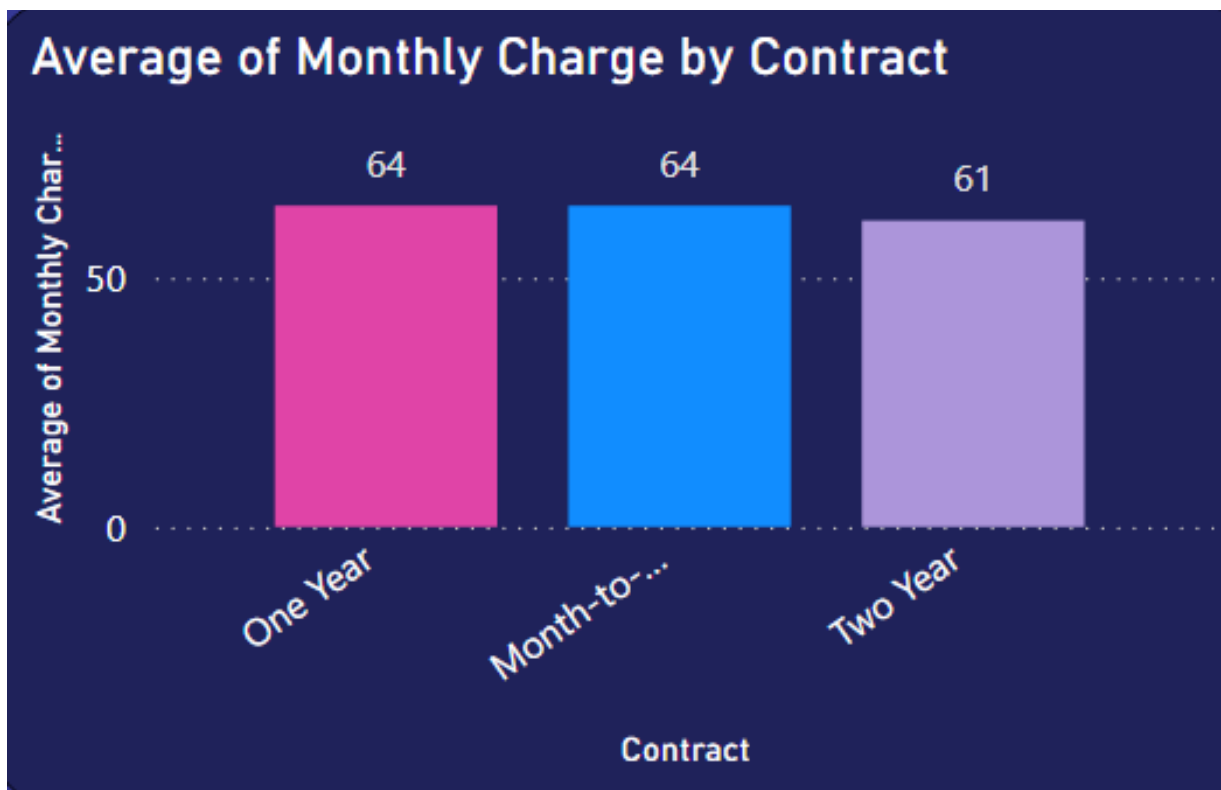
5.4 Revenue and Charges



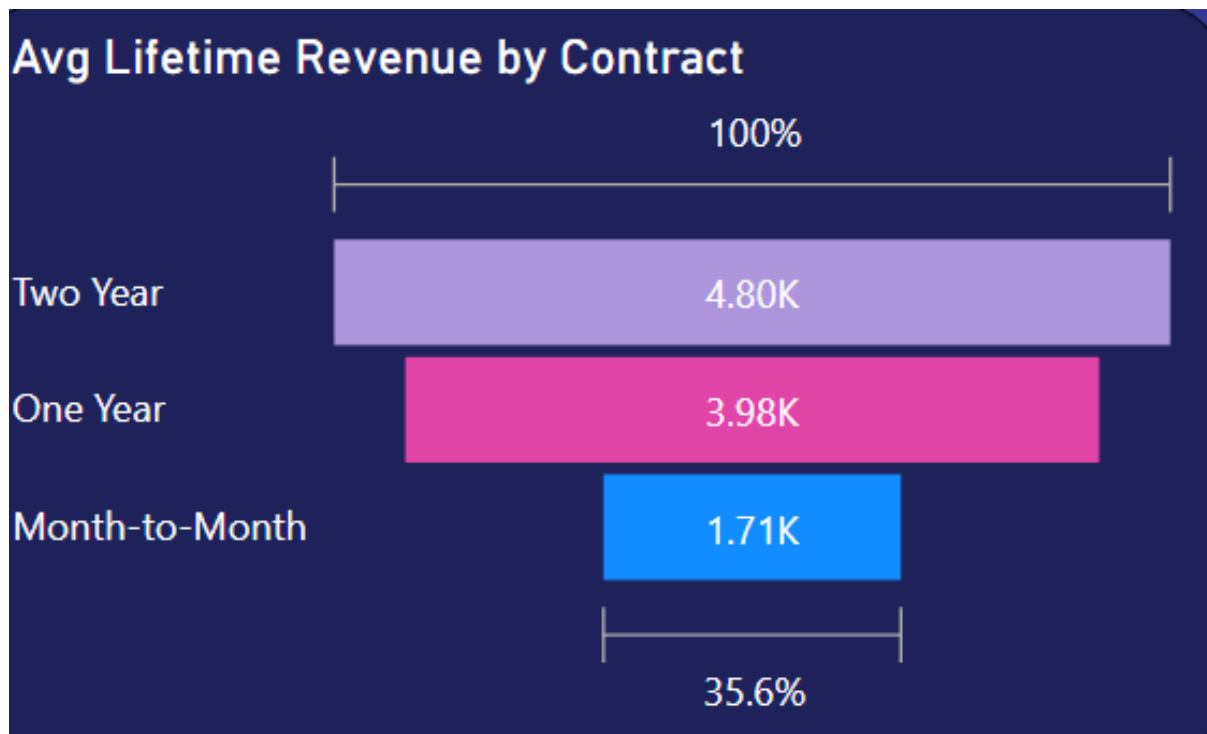
This dashboard shows the financial impact of churn. The company lost \$3.68M in revenue. Churned customers were generally higher-paying customers, so the company is losing many of its valuable customers.



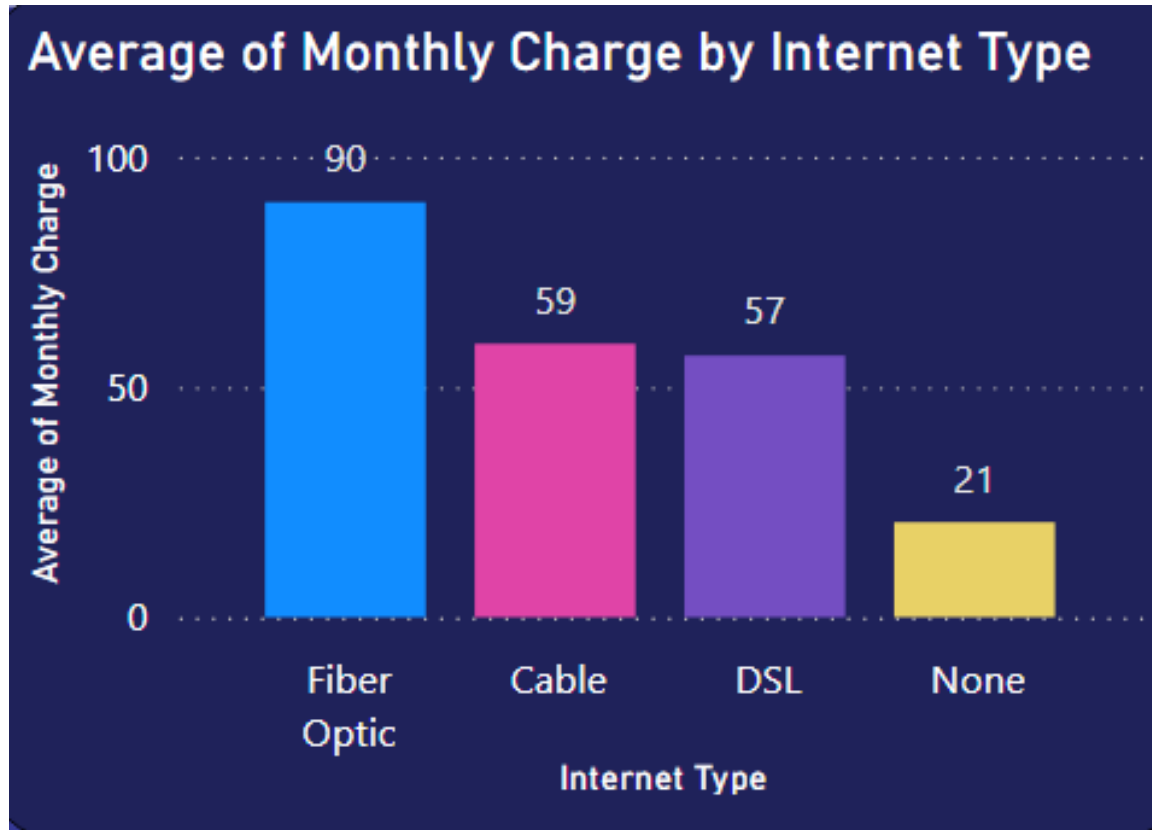
The company generated a total revenue of \$21.37M and lost \$3.68M in revenue from customers who left, equal to 17.24% of total revenue which indicates a significant financial issue. Customers who stayed paid an average monthly charge of \$61.74.



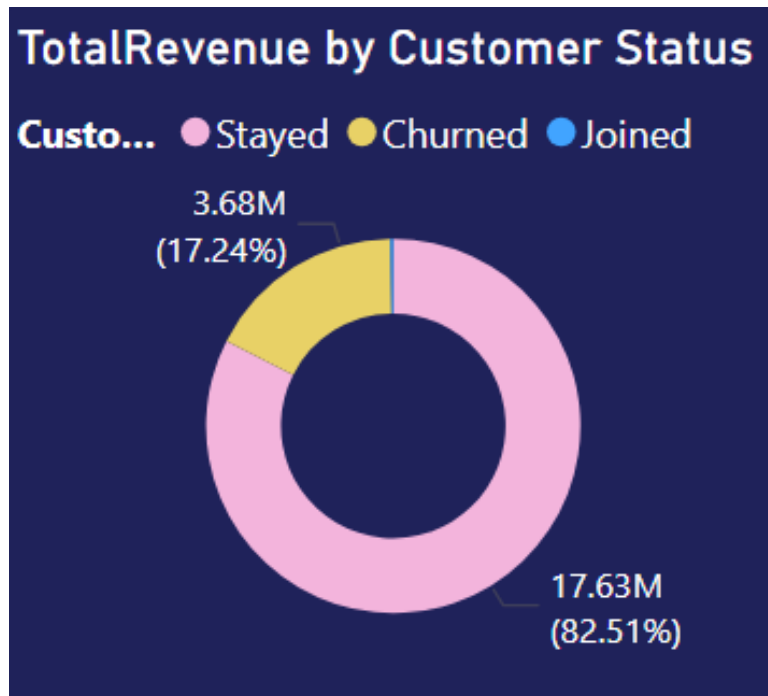
Monthly charges are similar across all contract types around \$61-\$64, showing that customer commitment matters more than pricing.



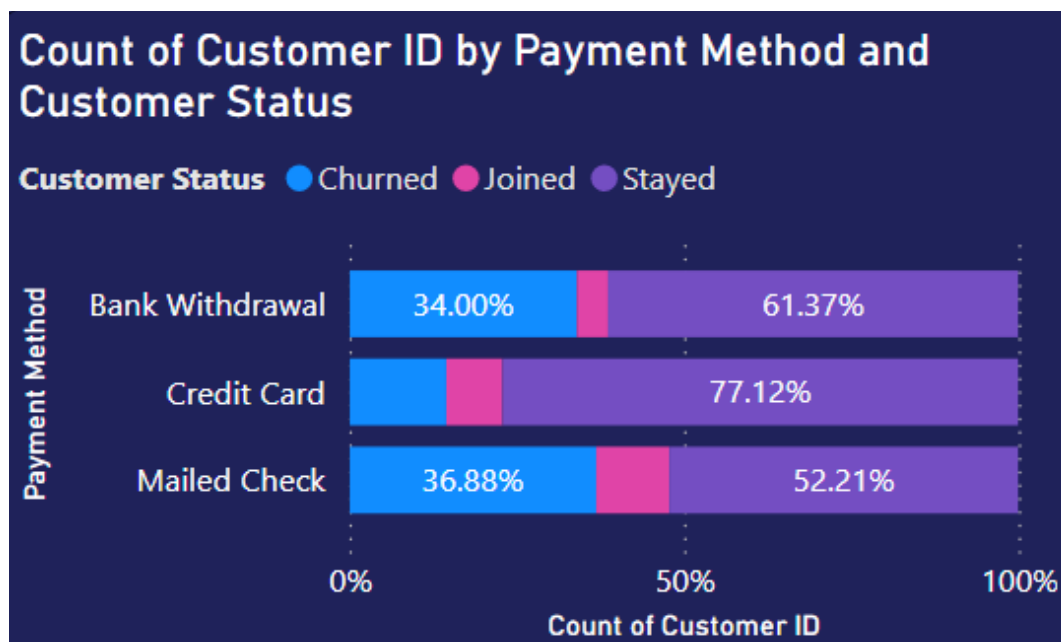
Two-year contract customers generate much higher lifetime revenue than month-to-month customers, making them more valuable to the business.

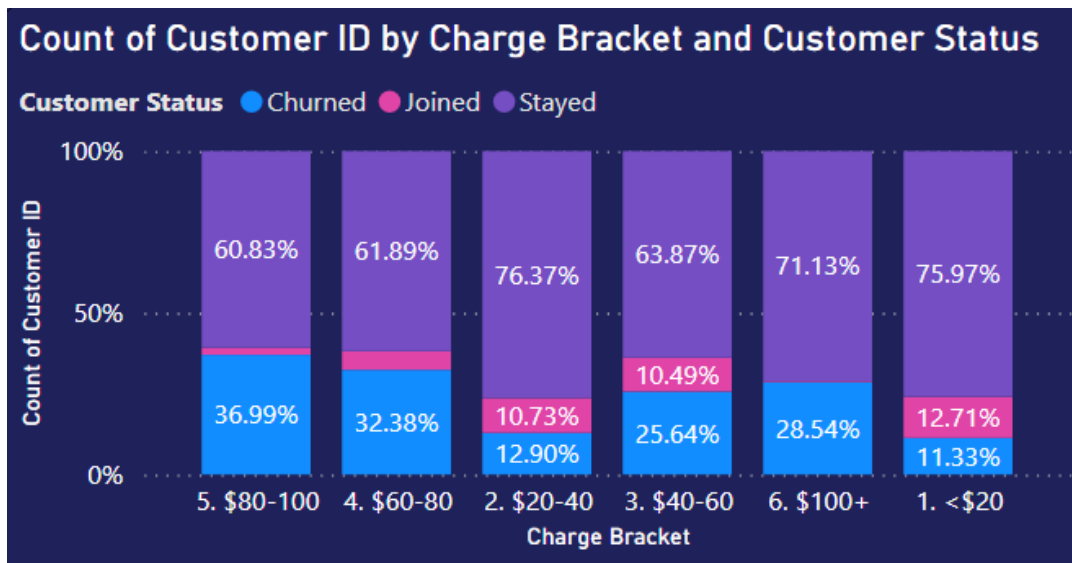


Fiber optic customers pay the highest monthly charges but also have the highest churn, suggesting dissatisfaction with value or service quality. This should be further investigated to increase satisfaction of customers.



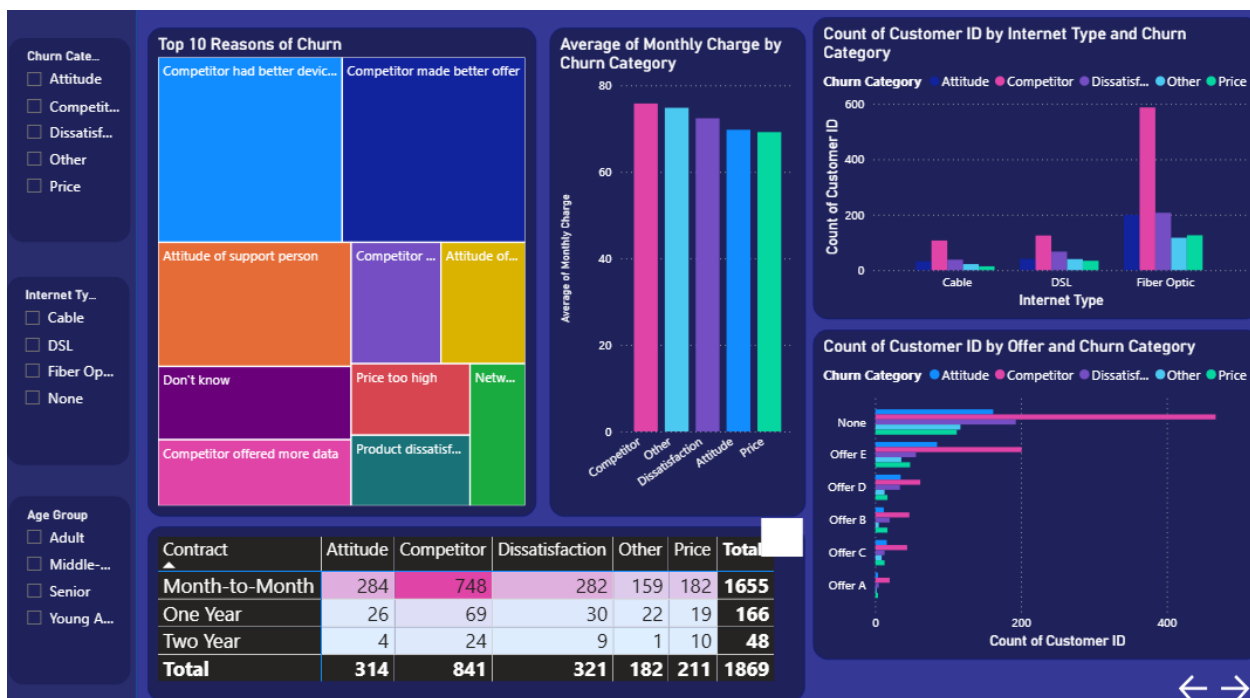
Most company revenue comes from retained customers, while churned customers account for a smaller share.



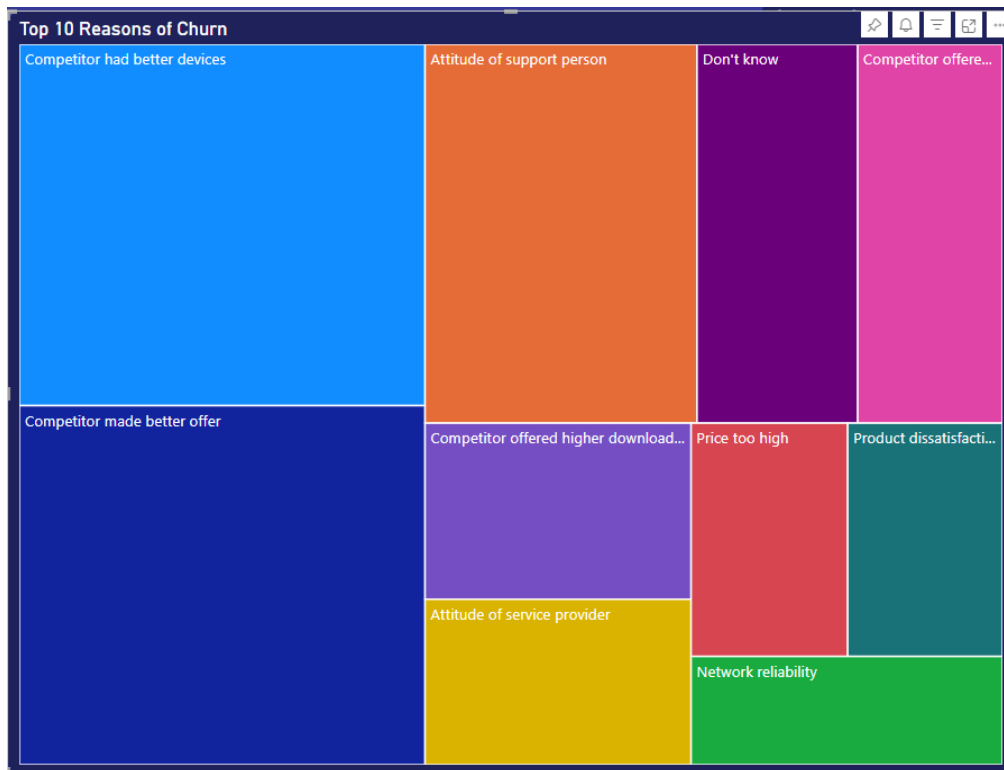


Customers using credit cards are more likely to stay, while customers paying by mailed checks show higher churn. Customers paying \$80-100 monthly show the highest churn at 36.99% while the lowest churn is in the <\$20 bracket at 11.33% suggesting both ends of the pricing ranges are at risk.

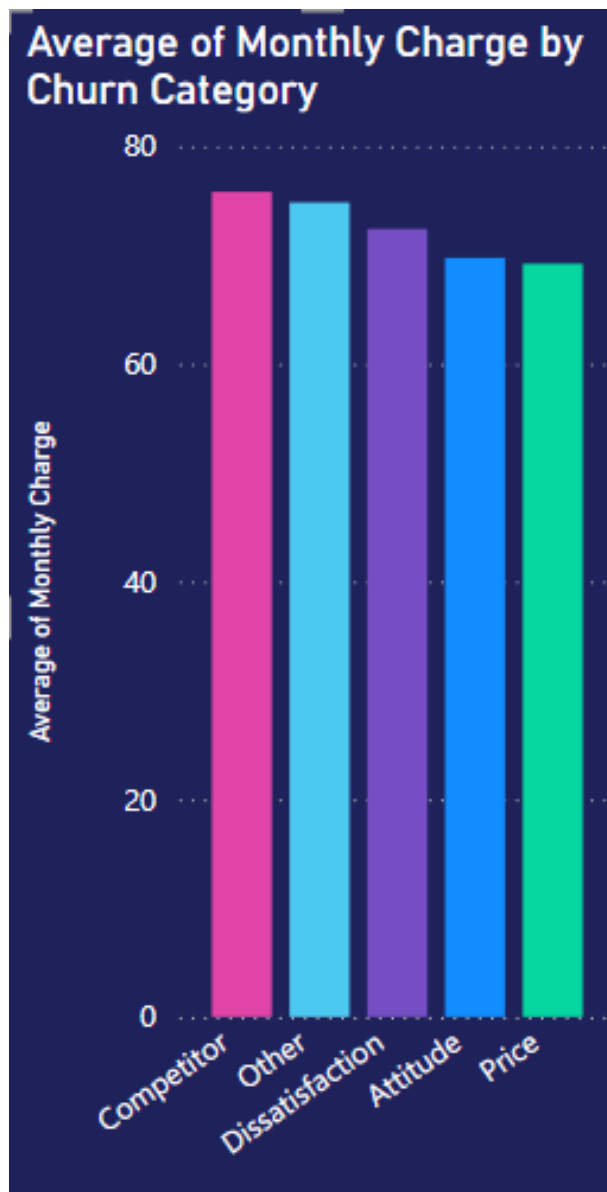
5.5 Churn Reasons



This dashboard explains why customers left the company. Competitors offering better devices and deals were the main reasons for churn. Most churned customers were on month-to-month contracts and had not received retention offers.



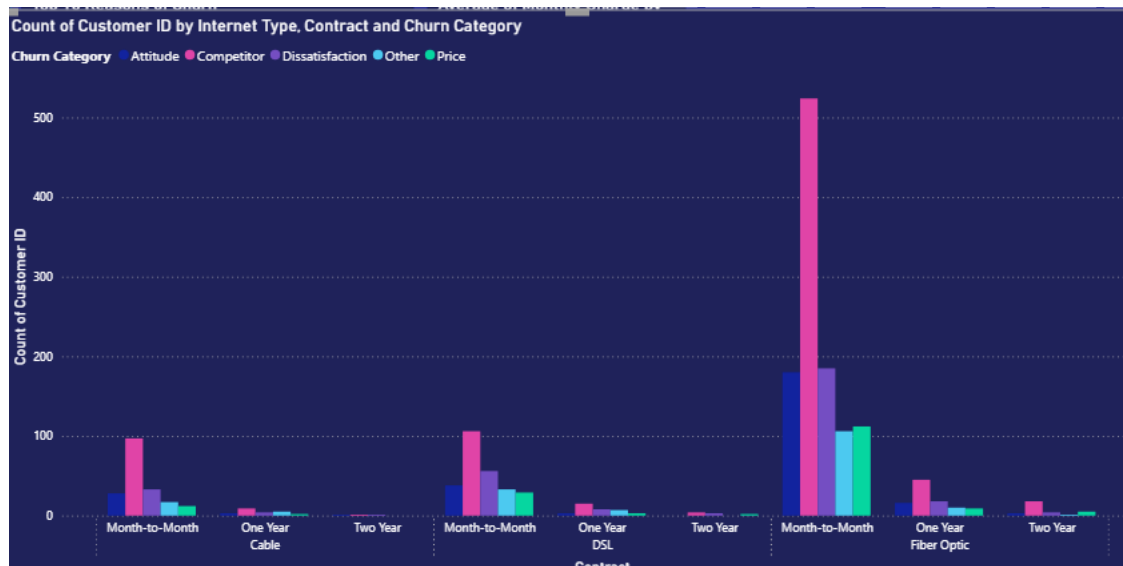
The biggest reasons for churn were competitors offering better devices and better deals, showing strong competitive pressure.



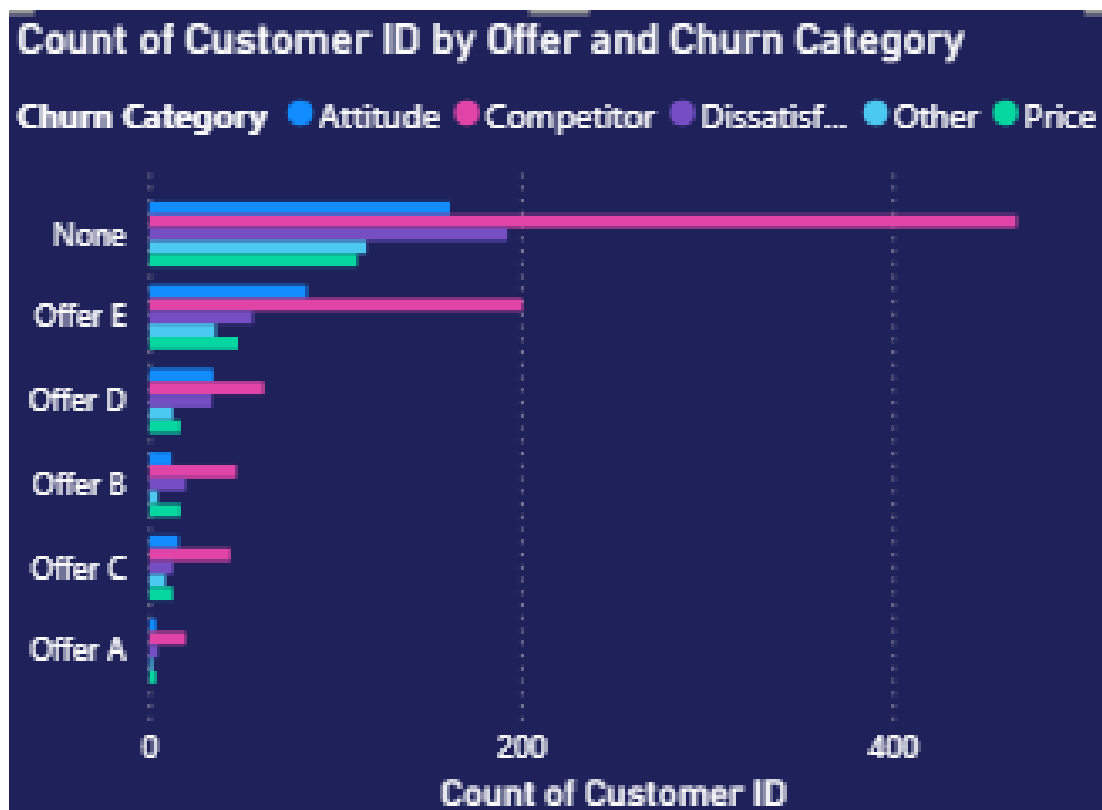
Customers who left because of competitors had the highest monthly charges about \$75, meaning the company is losing many high-value customers.

Contract	Attitude	Competitor	Dissatisfaction	Other	Price	Total
Month-to-Month	284	748	282	159	182	1655
One Year	26	69	30	22	19	166
Two Year	4	24	9	1	10	48
Total	314	841	321	182	211	1869

Month-to-month customers account for 1,655 churned customers with 748 specifically lost to competitors, while one-year and two-year contract customers churned far less but still high compared to other categories.



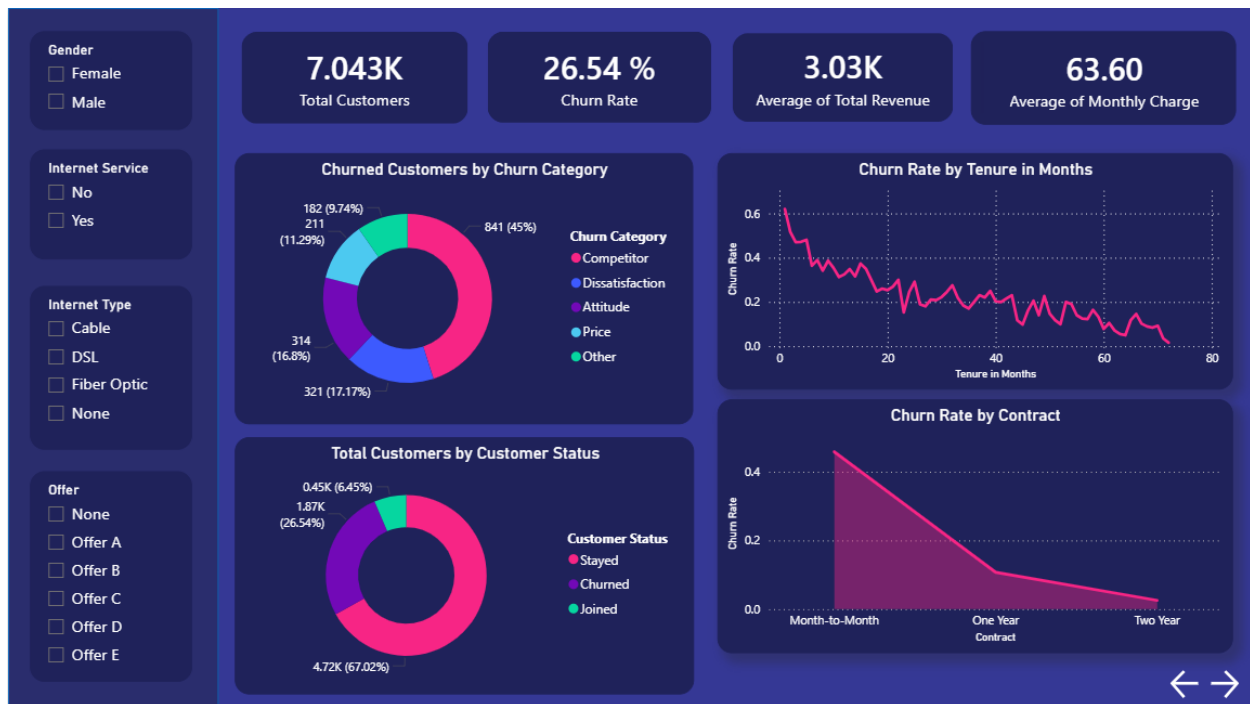
Fiber optic customers with month-to-month contracts had the highest competitor-related churn over 500, which makes it the most risky customer group.



Customers who received no offer had the highest churn to competitors over 400, while Offer A showed the lowest competitor churn and performed best.

6. Data Story Telling

6.1 Overview



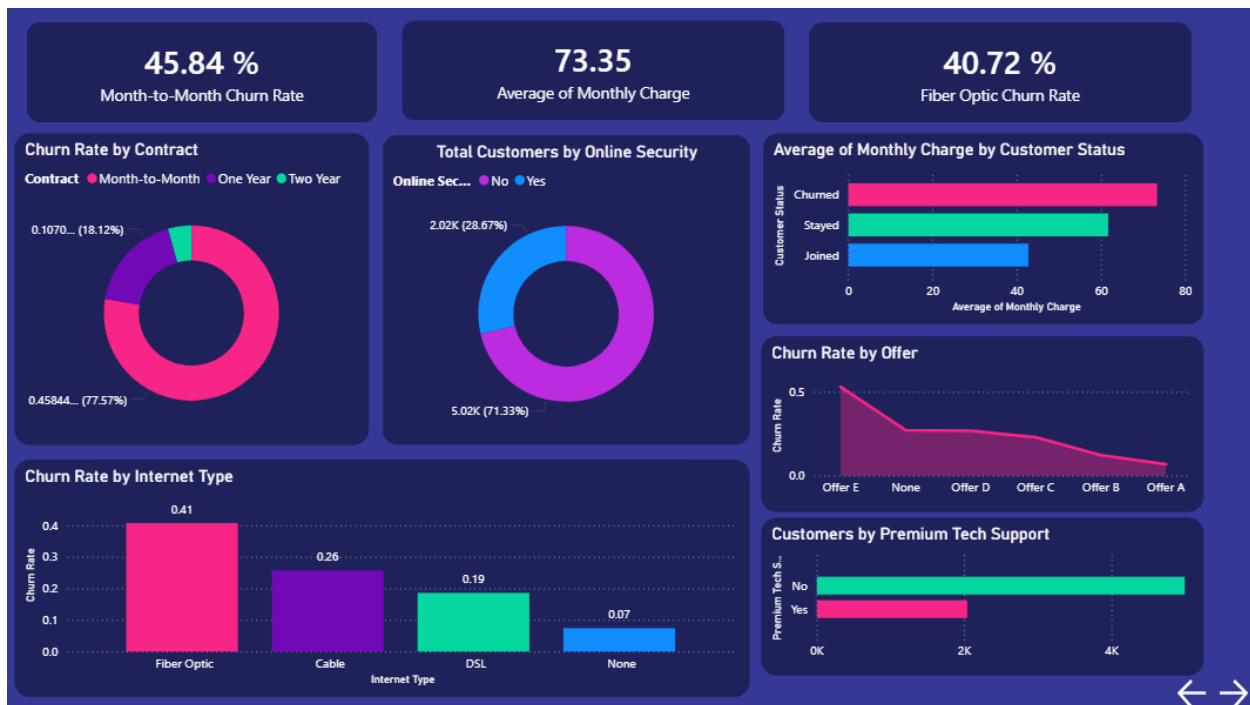
Our analysis shows us that more than quarter of our customers are leaving, indicating that 1 in 4 customers churn. The main driving factor of this is competitors causing churn at 45%.

6.2 Customer Demographic Analysis I and II



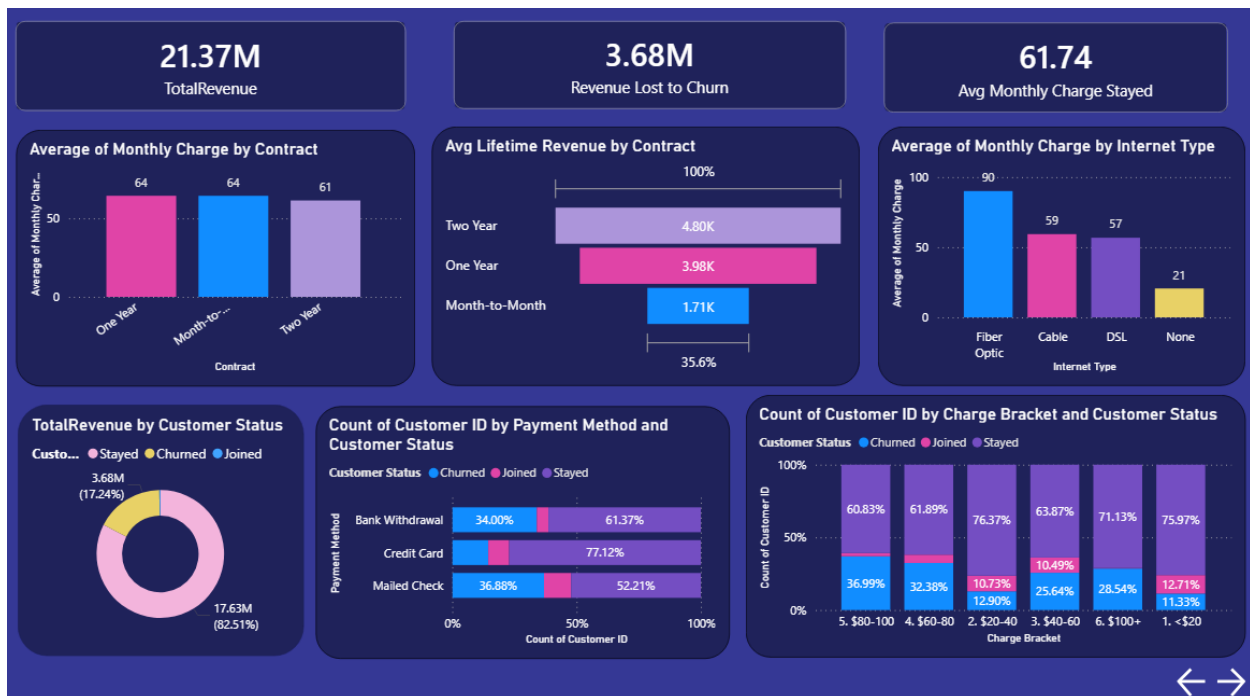
Customers most likely to churn are seniors, unmarried customers, and customers without dependents. These groups appear to have weaker loyalty to the service.

6.3 Services and Subscription



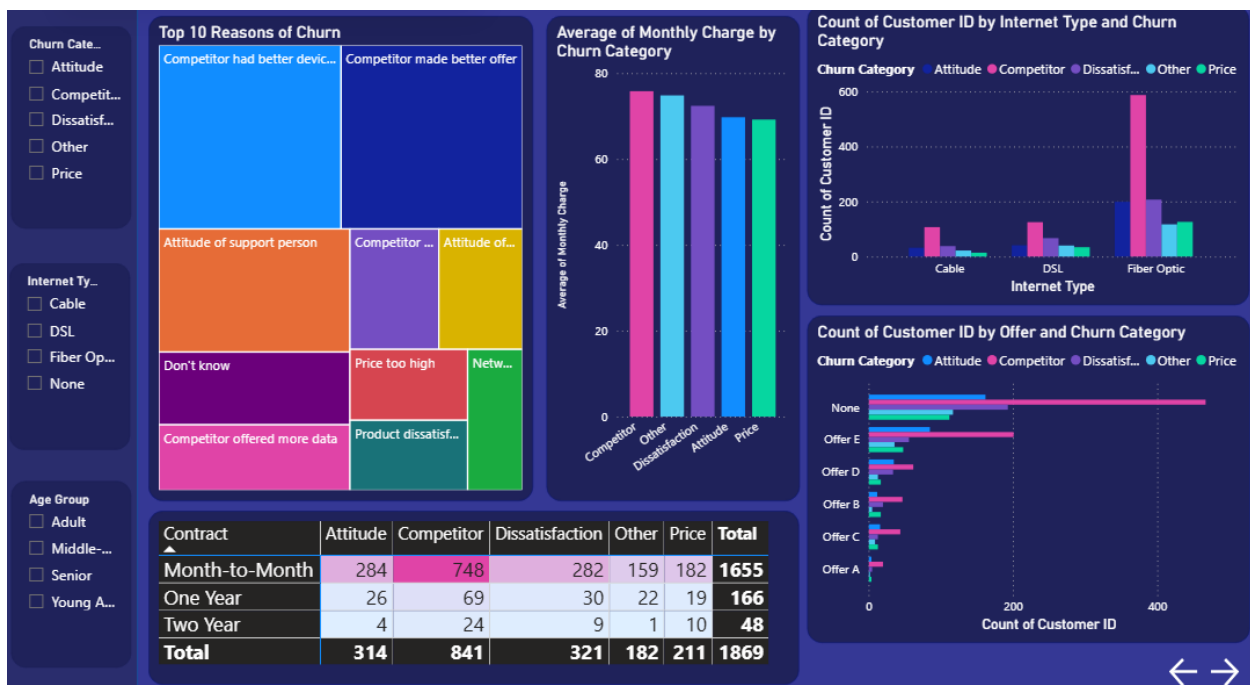
Month-to-month contract customers have the highest churn rate at 45.84%, while two-year contract customers have the lowest. Fiber optic users also show high churn despite paying the highest monthly charges. Customers without add-on services are more likely to leave.

6.4 Revenue and Charges



Customer churn has caused the company to lose \$3.68M in revenue. Churned customers paid higher monthly charges on average, showing that the company is losing many high-value customers.

6.5 Churn Reasons



Most customers left because competitors offered better devices and better deals. Customers who received no retention offer had the highest churn, while Offer A was the most successful in keeping customers.

6.6 Recommendations & Solutions

Based on our data story, we identified that the following key actions can significantly reduce churn:

- Promote Long-Term Contracts
- Use Offer A for High-Risk Customers: Provide Offer A to customers who are likely to leave, since it has the lowest churn rate among all offers.
- Focus on New Customers: Most churn happens in the first year, so introduce loyalty programs, customer support, and regular follow-ups for new customers. First-time discounts can also be offered.
- Strengthen Competitive Offers: Many customers left because competitors offered better devices and internet speeds. Regularly improve packages and promotions to stay competitive.
- Since additional services, particularly premium tech support and online security, improve customer retention, free trials can be offered for these services to encourage adoption.
- Service bundles can also be offered at discounted rates.
- 16.8% of churned customers cited attitude as a major reason for leaving. This highlights the need for customer service training for the telecom staff.
- Customers who refer others to the service could be rewarded with special offers to incentivise word-of-mouth marketing
- The majority of the telecom service users are middle-aged. Targeted campaigns should focus on retaining older customers, as data shows that senior customers are less likely to have premium tech support and tend to churn at higher rates.
- Since chi-square results revealed that payment method significantly impacts churn, targeted campaigns can be launched for check-paying customers, offering a small monthly discount or a one-time credit for switching to automatic bank withdrawal or credit card payment.

